

The Performance of Feedback Control Systems

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PREVIEW

The ability to adjust the transient and steady-state response of a control system is a beneficial outcome of the design of control systems. In this chapter, we introduce the time-domain performance specifications and use key input signals to test the response of the control system. The correlation between the system performance and the location of the transfer function poles and zeros is discussed. We develop relationships between the performance specifications and the natural frequency and damping ratio for second-order systems. Relying on the notion of dominant poles, we can extrapolate the ideas associated with second-order systems to those of higher order. The concept of a performance index is also considered. We present a set of quantitative performance indices that adequately represent the performance of the control system. The chapter concludes with a performance analysis of the Sequential Design Example: Disk Drive Read System.

DESIRED OUTCOMES

Upon completion of Chapter 5, students should:

- ❑ Be aware of key test signals used in controls and of the resulting transient response characteristics of second-order systems to test signal inputs.
- ❑ Recognize the direct relationship between the pole locations of second-order systems and the transient response.
- ❑ Be familiar with the design formulas that relate the second-order pole locations to percent overshoot, settling time, rise time, and time to peak.
- ❑ Be aware of the impact of a zero and a third pole on the second-order system response.
- ❑ Gain a sense of optimal control as measured with performance indices.

5.1 INTRODUCTION

The ability to adjust the transient and steady-state performance is a distinct advantage of feedback control systems. To analyze and design a control system, we must define and measure its performance. Based on the desired performance of the control system, the controller parameters are adjusted to provide the desired response. Because control systems are inherently dynamic, their performance is usually specified in terms of both the transient response and the steady-state response. The **transient response** is the response that disappears with time. The **steady-state response** is the response that exists for a long time following an input signal initiation.

The **design specifications** for control systems normally include several time-response indices for a specified input command, as well as a desired steady-state accuracy. In the course of the design, the specifications are often revised to effect a compromise. Therefore, specifications are seldom a rigid set of requirements, but rather an attempt to quantify the desired performance. The effective compromise and adjustment of specifications are graphically illustrated in Figure 5.1. The parameter p may minimize the performance measure M_2 if we select p as a very small value. However, this results in large measure M_1 , an undesirable situation. If the performance measures are equally important, the crossover point at $p_{\min}$ provides the best compromise. This type of compromise is often encountered in feedback control system design. It is clear that if the original specifications called for both M_1 and M_2 to be minimized, the specifications could not be simultaneously met; they would then have to be altered to allow for the compromise resulting with $p_{\min}$ [1, 10, 15, 20].

The specifications, which are stated in terms of the measures of performance, indicate the quality of the system to the designer. In other words, the performance measures help to answer the question, How well does the system perform the task for which it was designed?

5.2 TEST INPUT SIGNALS

Time-domain performance specifications are important indices because control systems are inherently time-domain systems. The transient response is of prime interest for control system designers. It is necessary to determine initially whether

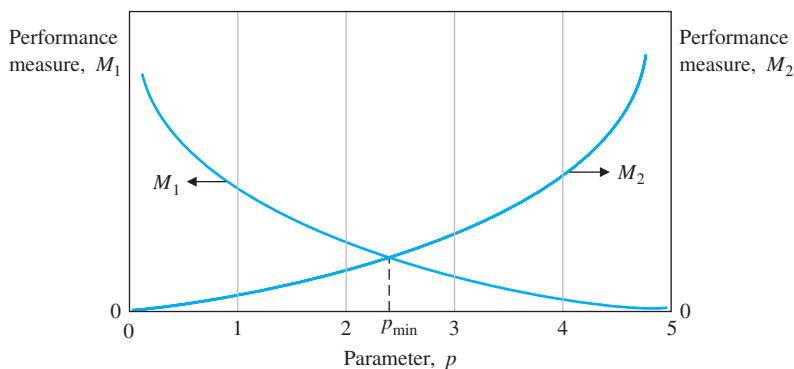


FIGURE 5.1
Two performance measures versus parameter p .

Table 5.1 Test Signal Inputs

Test Signal	$r(t)$	$R(s)$
Step	$r(t) = A, t > 0$ $= 0, t < 0$	$R(s) = A/s$
Ramp	$r(t) = At, t > 0$ $= 0, t < 0$	$R(s) = A/s^2$
Parabolic	$r(t) = At^2, t > 0$ $= 0, t < 0$	$R(s) = 2A/s^3$

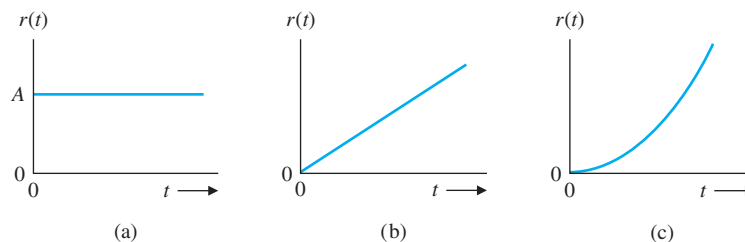
the system is stable; we can achieve this goal by using the techniques of ensuing chapters. If the system is stable, the response to a specific input signal will provide several measures of the performance. However, because the actual input signal of the system is usually unknown, a standard **test input signal** is normally chosen. This approach is quite useful because there is a reasonable correlation between the response of a system to a standard test input and the system's ability to perform under normal operating conditions. Furthermore, using a standard input allows the designer to compare several competing designs. Many control systems experience input signals that are very similar to the standard test signals.

The standard test input signals commonly used are the step input, the ramp input, and the parabolic input. These inputs are shown in Figure 5.2. The equations representing these test signals are given in Table 5.1, where the Laplace transform can be obtained by using Table 2.3 and a more complete list of Laplace transform pairs can be found at the MCS website. The ramp signal is the integral of the step input, and the parabola is the integral of the ramp input. A **unit impulse** function is also useful for test signal purposes. The unit impulse is based on a rectangular function

$$f_{\epsilon}(t) = \begin{cases} 1/\epsilon, & -\frac{\epsilon}{2} \leq t \leq \frac{\epsilon}{2}; \\ 0, & \text{otherwise,} \end{cases}$$

where $\epsilon > 0$. As ϵ approaches zero, the function $f_{\epsilon}(t)$ approaches the unit impulse function $\delta(t)$, which has the following properties

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t - a)g(t) dt = g(a). \quad (5.1)$$

**FIGURE 5.2**

Test input signals:
(a) step, (b) ramp,
and (c) parabolic.

The impulse input is useful when we consider the convolution integral for the output $y(t)$ in terms of an input $r(t)$, which is written as

$$y(t) = \int_{-\infty}^t g(t - \tau)r(\tau) d\tau = \mathcal{L}^{-1}\{G(s)R(s)\}. \quad (5.2)$$

The relationship in Equation (5.2) represents the open-loop input–output relationship of a system $G(s)$. If the input is a unit impulse function, we have

$$y(t) = \int_{-\infty}^t g(t - \tau)\delta(\tau) d\tau. \quad (5.3)$$

The integral has a value only at $\tau = 0$; therefore,

$$y(t) = g(t),$$

the impulse response of the system $G(s)$. The impulse response test signal can often be used for a dynamic system by subjecting the system to a large-amplitude, narrow-width pulse of area A .

The standard test signals are of the general form

$$r(t) = t^n, \quad (5.4)$$

and the Laplace transform is

$$R(s) = \frac{n!}{s^{n+1}}. \quad (5.5)$$

Hence, the response to one test signal may be related to the response of another test signal of the form of Equation (5.4). The step input signal is the easiest to generate and evaluate and is usually chosen for performance tests.

Consider the response of a system $G(s)$ for a unit step input, $R(s) = 1/s$, when

$$G(s) = \frac{9}{s + 10}.$$

Then the output is

$$Y(s) = \frac{9}{s(s + 10)},$$

the response during the transient period is

$$y(t) = 0.9(1 - e^{-10t}),$$

and the steady-state response is

$$y(\infty) = 0.9.$$

If the error is $E(s) = R(s) - Y(s)$, then the steady-state error is

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{s + 1}{s + 10} = 0.1.$$

5.3 PERFORMANCE OF SECOND-ORDER SYSTEMS

Let us consider a single-loop second-order system and determine its response to a unit step input. A closed-loop feedback control system is shown in Figure 5.3. The closed-loop transfer function is

$$Y(s) = \frac{G(s)}{1 + G(s)} R(s). \quad (5.6)$$

We may rewrite Equation (5.6) as

$$Y(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s). \quad (5.7)$$

With a unit step input, we obtain

$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}, \quad (5.8)$$

from which it follows that

$$y(t) = 1 - \frac{1}{\beta} e^{-\zeta\omega_n t} \sin(\omega_n \beta t + \theta), \quad (5.9)$$

where $\beta = \sqrt{1 - \zeta^2}$, $\theta = \cos^{-1} \zeta$, and $0 < \zeta < 1$. The response of this second-order system for various values of the damping ratio ζ is shown in Figure 5.4. As ζ decreases, the closed-loop poles approach the imaginary axis, and the response becomes increasingly oscillatory.

The Laplace transform of the unit impulse is $R(s) = 1$, and therefore the output for an impulse is

$$Y(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (5.10)$$

The response for an impulse function input is then

$$y(t) = \frac{\omega_n}{\beta} e^{-\zeta\omega_n t} \sin(\omega_n \beta t), \quad (5.11)$$

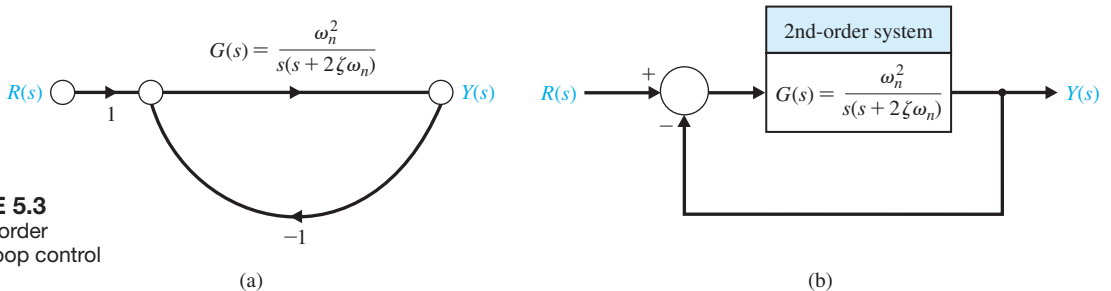


FIGURE 5.3
Second-order
closed-loop control
system.

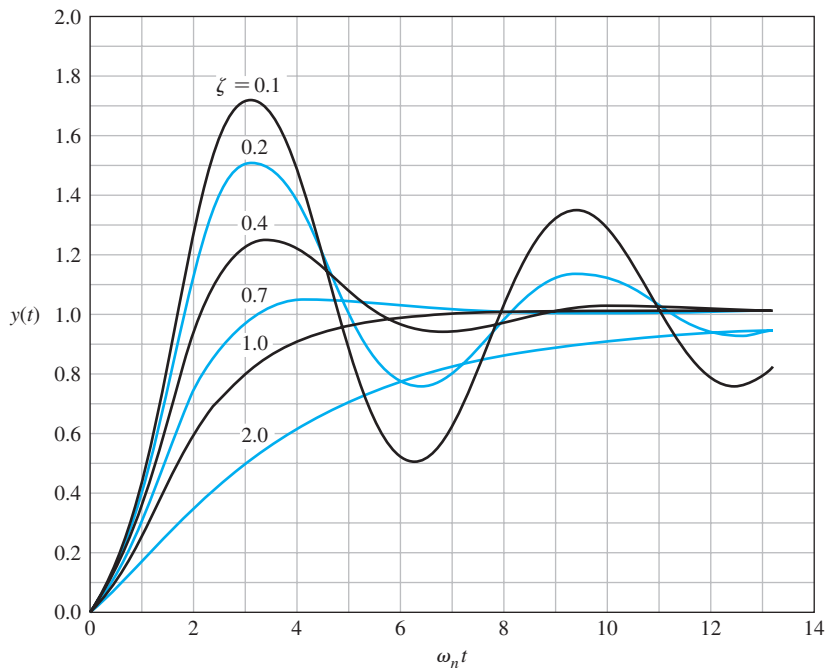


FIGURE 5.4
Transient response
of a second-order
system for a step
input.

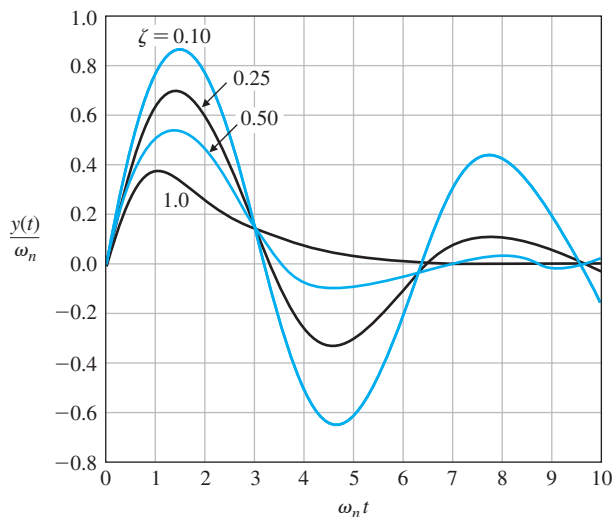


FIGURE 5.5
Response of a
second-order
system for an
impulse input.

which is the derivative of the response to a step input. The impulse response of the second-order system is shown in Figure 5.5 for several values of the damping ratio ζ .

Standard performance measures are often defined in terms of the step response of the closed-loop system as shown in Figure 5.6. The swiftness of the response is measured by the **rise time** T_r and the **peak time** T_p . For underdamped systems with an overshoot, the 0–100% rise time is a useful index. If the system is overdamped, then the peak time is not defined, and the 10–90% rise time T_{r1} is normally used.

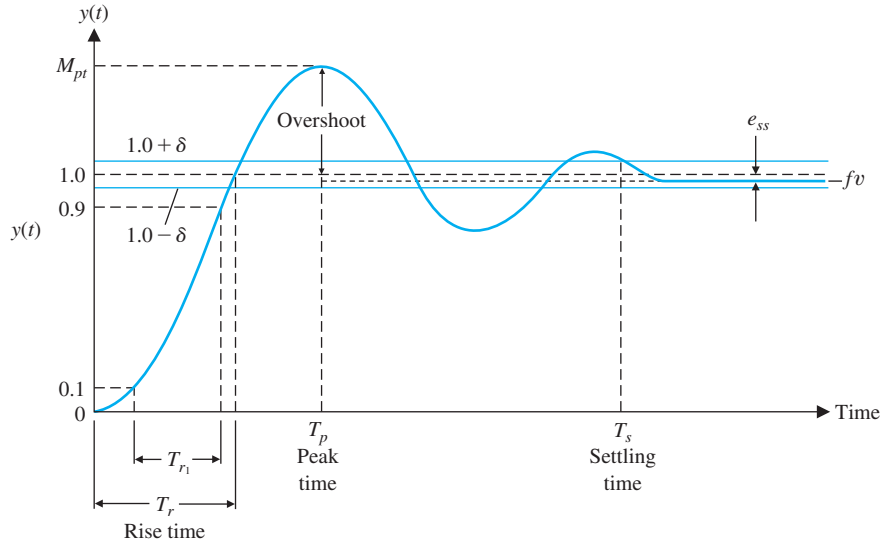


FIGURE 5.6
Step response of
a second-order
system.

The similarity with which the actual response matches the step input is measured by the percent overshoot and settling time T_s . The **percent overshoot** is defined as

$$P.O. = \frac{M_{pt} - fv}{fv} \times 100\% \quad (5.12)$$

for a unit step input, where M_{pt} is the peak value of the time response, and fv is the final value of the response. Normally, fv is the magnitude of the input, but many systems have a final value significantly different from the desired input magnitude. For the system with a unit step represented by Equation (5.8), we have $fv = 1$.

The **settling time**, T_s , is defined as the time required for the system to settle within a certain percentage δ of the input amplitude. This band of $\pm\delta$ is shown in Figure 5.6. For the second-order system with closed-loop damping constant $\zeta\omega_n$ and a response described by Equation (5.9), we seek to determine the time T_s for which the response remains within 2% of the final value. This occurs approximately when

$$e^{-\zeta\omega_n T_s} < 0.02,$$

or

$$\zeta\omega_n T_s \cong 4.$$

Therefore, we have

$$T_s = 4\tau = \frac{4}{\zeta\omega_n}. \quad (5.13)$$

Hence, we define the settling time as four time constants (that is, $\tau = 1/\zeta\omega_n$) of the dominant roots of the characteristic equation. The steady-state error of the system may be measured on the step response of the system as shown in Figure 5.6.

The transient response of the system may be described in terms of two factors:

1. The swiftness of response, as represented by the rise time and the peak time.
2. The closeness of the response to the desired response, as represented by the overshoot and settling time.

As it turns out, these are often contradictory requirements; thus, a compromise must be obtained. To obtain an explicit relation for M_{pt} and T_p as a function of ζ , one can differentiate Equation (5.9) and set it equal to zero yielding

$$\dot{y}(t) = \frac{\omega_n}{\beta} e^{-J\omega_n t} \sin(\omega_n \beta t) = 0,$$

which is equal to zero when $\omega_n \beta t = n\pi$, where $n = 0, 1, 2, \dots$. The first nonzero time this is equal to zero is when $n = 1$. Thus, we find that the peak time relationship for this second-order system is

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}, \quad (5.14)$$

and the peak response is

$$M_{pt} = 1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}}. \quad (5.15)$$

Therefore, the percent overshoot is

$$P.O. = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}. \quad (5.16)$$

The percent overshoot versus the damping ratio, ζ , is shown in Figure 5.7. Also, the normalized peak time, $\omega_n T_p$, is shown versus the damping ratio, ζ , in Figure 5.7. Upon inspection of Figure 5.7, we see that we are confronted with a necessary compromise between the swiftness of response and the allowable percent overshoot.

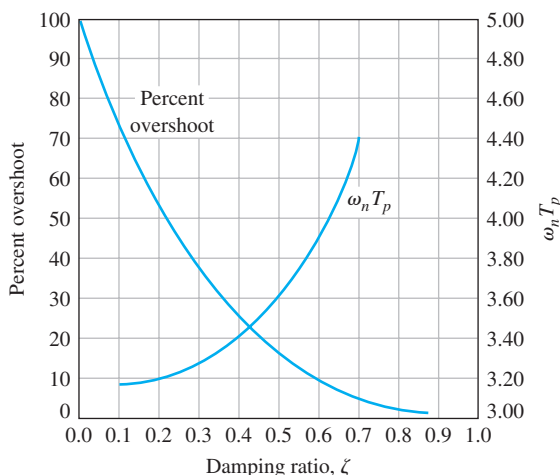


FIGURE 5.7 Percent overshoot and normalized peak time versus damping ratio ζ for a second-order system (Equation 5.8).

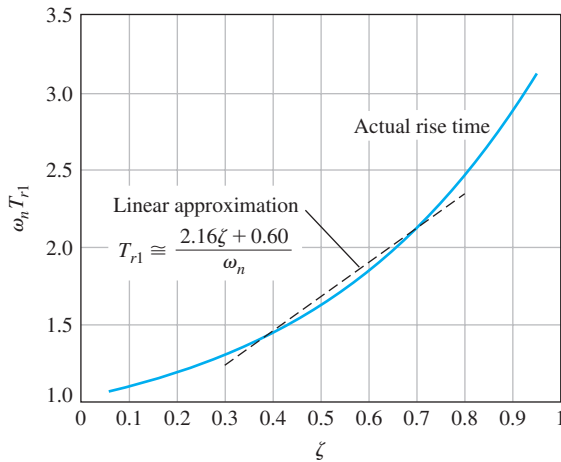


FIGURE 5.8
Normalized rise time, T_{r1} , versus ζ for a second-order system.

The swiftness of step response can be measured as the time it takes to rise from 10% to 90% of the magnitude of the step input. This is the definition of the rise time, T_{r1} , shown in Figure 5.6. The normalized rise time, $\omega_n T_{r1}$, versus ζ ($0.05 \leq \zeta \leq 0.95$) is shown in Figure 5.8. Although it is difficult to obtain exact analytic expressions for T_{r1} , we can utilize the linear approximation

$$T_{r1} = \frac{2.16\zeta + 0.60}{\omega_n}, \quad (5.17)$$

which is accurate for $0.3 \leq \zeta \leq 0.8$. This linear approximation is shown in Figure 5.8.

The swiftness of a response to a step input as described by Equation (5.17) is dependent on ζ and ω_n . For a given ζ , the response is faster for larger ω_n , as shown in Figure 5.9. Note that the percent overshoot is independent of ω_n .

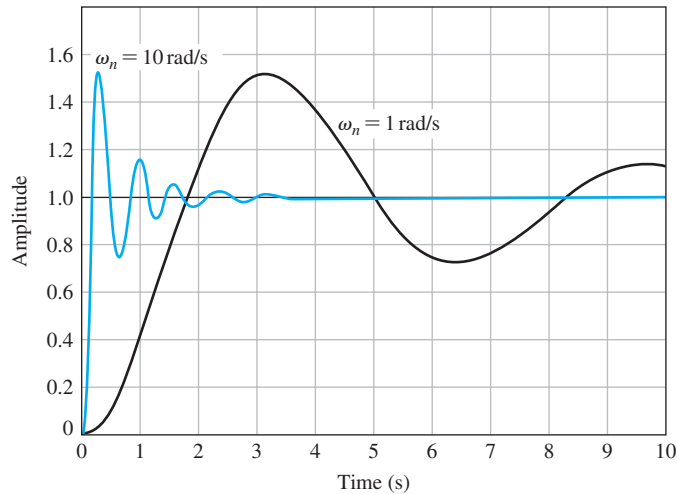


FIGURE 5.9
The step response for $\zeta = 0.2$ for $\omega_n = 1$ and $\omega_n = 10$.

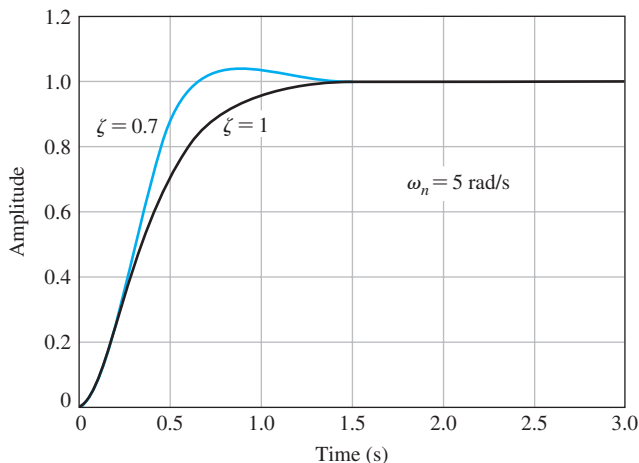


FIGURE 5.10
The step response
for $\omega_n = 5$ with
 $\zeta = 0.7$ and $\zeta = 1$.

For a given ω_n , the response is faster for lower ζ , as shown in Figure 5.10. The swiftness of the response, however, will be limited by the overshoot that can be accepted.

5.4 EFFECTS OF A THIRD POLE AND A ZERO ON THE SECOND-ORDER SYSTEM RESPONSE

The curves presented in Figure 5.7 are exact only for the second-order system of Equation (5.8). However, they provide important information because many systems possess a dominant pair of roots and the step response can be estimated by utilizing Figure 5.7. This approach, although an approximation, avoids the evaluation of the inverse Laplace transformation in order to determine the percent overshoot and other performance measures. For example, for a third-order system with a closed-loop transfer function

$$T(s) = \frac{1}{(s^2 + 2\zeta s + 1)(\gamma s + 1)}, \quad (5.18)$$

the s -plane diagram is shown in Figure 5.11. This third-order system is normalized with $\omega_n = 1$. The performance (as indicated by the percent overshoot, $P.O.$, and the settling time, T_s), is adequately represented by the second-order system approximation when [4]

$$|1/\gamma| \geq 10|\zeta\omega_n|.$$

In other words, the response of a third-order system can be approximated by the **dominant roots** of the second-order system as long as the real part of the dominant roots is less than one tenth of the real part of the third root [15, 20].

Consider the third-order system

$$T(s) = \frac{1}{(s^2 + 2\zeta\omega_n s + 1)(\gamma s + 1)}$$

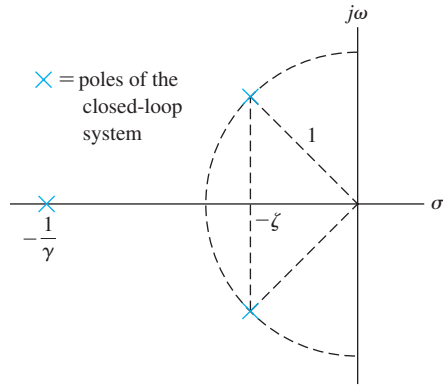


FIGURE 5.11
An s-plane diagram
of a third-order
system.

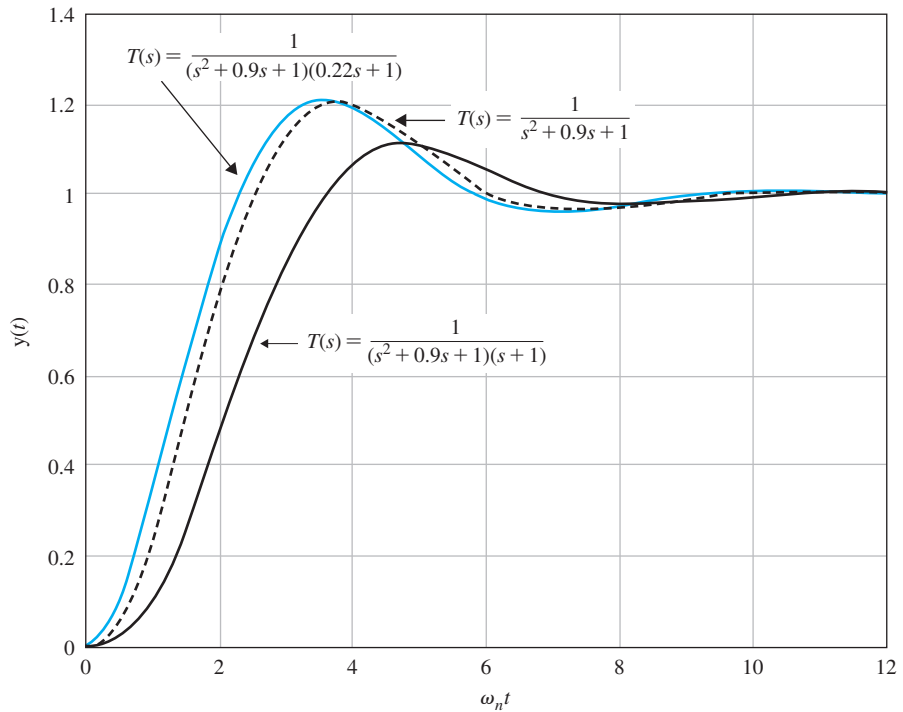


FIGURE 5.12
Comparison of two
third-order systems
with a second-order
system (dashed
line) illustrating
the concept
of dominant
poles when
 $|1/\gamma| \geq 10\zeta\omega_n$.

where $\omega_n = 1.0$, $\zeta = 0.45$, and $\gamma = 1.0$. In this case, $|1/\gamma| \not\geq 10\zeta\omega_n$. The system poles are at $s_{1,2} = -0.45 \pm 0.89i$ and $s_3 = -1.0$. As illustrated in Figure 5.12, the percent overshoot is $P.O. = 10.9\%$, the settling time (to within 2% of the final value) is $T_s = 8.84$ s, and the rise time $T_{r1} = 2.16$ s. Suppose that we have another third-order system with $\omega_n = 1.0$, $\zeta = 0.45$, and $\gamma = 0.22$. Then the system poles are at $s_{1,2} = -0.45 \pm 0.89i$ (the same as the first system) and $s_3 = -4.5$. In this case, $|1/\gamma| \geq 10\zeta\omega_n$ and the complex poles pair are the dominant poles. As illustrated in Figure 5.12, the percent overshoot is $P.O. = 20.0\%$, the settling time

is $T_s = 8.56$ s, and the rise time $T_{r1} = 1.6$ s. When the complex pair of poles are the dominant poles, we can create the second-order system approximation

$$\hat{T}(s) = \frac{1}{s^2 + 2\zeta\omega_n s + 1} = \frac{1}{s^2 + 0.9s + 1}$$

and we would expect the percent overshoot, settling time, and rise time to be $P.O. = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}} = 20.5\%$, $T_s = 4/\zeta\omega_n = 8.89$ s, and $T_{r1} = (2.16\zeta + 0.6)/\omega_n = 1.57$ s, respectively. In Figure 5.12, it is evident that for the third-order system satisfying the condition $|1/\gamma| \geq 10\zeta\omega_n$, the step response more closely matches the response of the second-order system, as expected.

The performance measures associated with the second-order system in Equation (5.10) are precise only for transfer functions without finite zeros. If the transfer function of a system possesses a finite zero and it is located relatively near the dominant complex poles, then the zero will materially affect the transient response of the system. In other words, the transient response of a system with one zero and two poles may be affected by the location of the zero [5]. Consider a system with the system transfer function

$$T(s) = \frac{(\omega_n^2/a)(s + a)}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

We can investigate the response of the system compared to a second-order system without the finite zero. Suppose that $\zeta = 0.45$ and let $a/\zeta\omega_n = 0.5, 1, 2$, and 10.0 . The resulting unit step responses are shown in Figure 5.13. As the ratio $a/\zeta\omega_n$ increases, the finite zero moves farther into the left half-plane and away from the poles, and the step response approaches the second-order system response, as expected.

The correlation of the time-domain response of a system with the s -plane location of the poles of the closed-loop transfer function is a key concept in understanding system performance in the closed-loop.

EXAMPLE 5.1 Parameter selection

A single-loop feedback control system is shown in Figure 5.14. We need to select the controller gain K and the parameter p so that the time-domain specifications are satisfied. The transient response to a unit step is specified to have a percent overshoot of $P.O. \leq 5\%$ and a settling time to within 2% of the final value of $T_s \leq 4$ s. For second-order systems, we know the relationship in Equation (5.16) between percent overshoot and ζ , and the relationship in Equation (5.13) between settling time and $\zeta\omega_n$. Solving for $P.O. \leq 5\%$ yields $\zeta \geq 0.69$ and solving for $T_s \leq 4$ s yields $\zeta\omega_n \geq 1$.

The region that will satisfy both time-domain requirements is shown on the s -plane of Figure 5.15. To meet the performance specifications, we can choose $\zeta = 0.707$ ($P.O. = 4.3\%$) and $\zeta\omega_n = 1$ ($T_s = 4$ s). Hence, the desired closed-loop

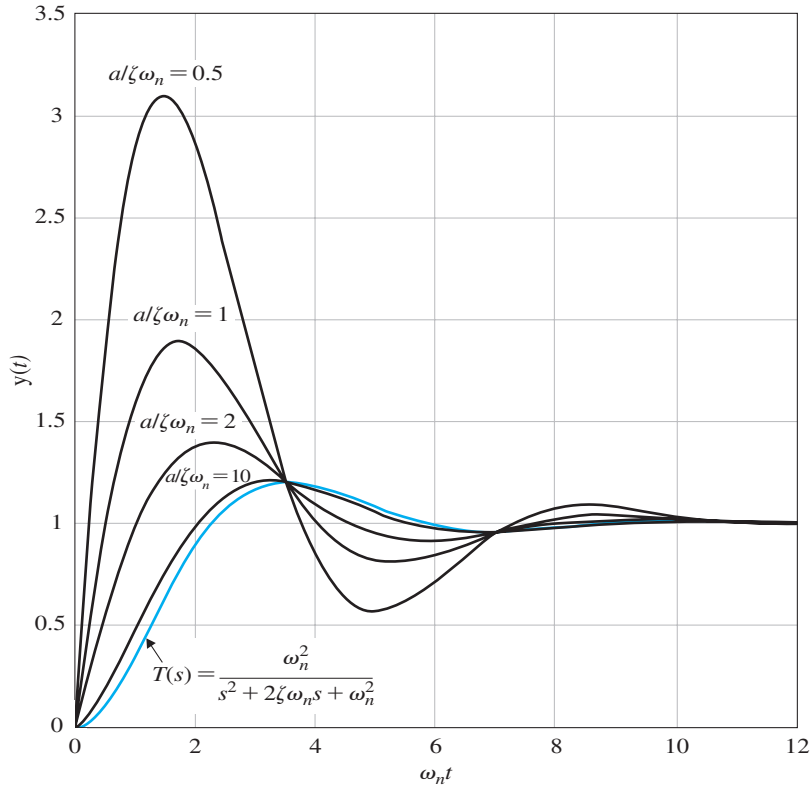


FIGURE 5.13
The response for the second-order transfer function with a zero for four values of the ratio $a/\zeta\omega_n = 0.5, 1, 2$, and 10.0 when $\zeta = 0.45$.

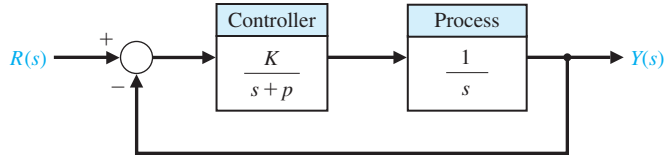


FIGURE 5.14
Single-loop feedback control system.

poles are $r_1 = -1 + j1$ and $\hat{r}_1 = -1 - j1$. Therefore, $\zeta = 1/\sqrt{2}$ and $\omega_n = 1/\zeta = \sqrt{2}$. The closed-loop transfer function is

$$T(s) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)} = \frac{K}{s^2 + ps + K} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

Solving for K and p yields $K = \omega_n^2 = 2$ and $p = 2\zeta\omega_n = 2$. Since this is exactly a second-order system of the form in Equation (5.7), the time-domain performance specifications will be precisely satisfied.

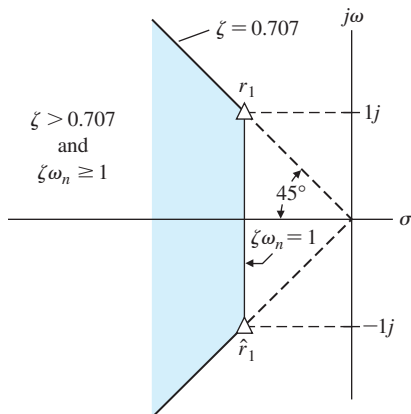


FIGURE 5.15
Specifications and
root locations on
the s-plane.

EXAMPLE 5.2 Impact of a zero and an additional pole

Consider a system with a closed-loop transfer function

$$\frac{Y(s)}{R(s)} = T(s) = \frac{\frac{\omega_n^2}{a}(s + a)}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(1 + \tau s)}.$$

Both the zero and the real pole may affect the transient response. If $a \gg \zeta\omega_n$ and $\tau \ll 1/\zeta\omega_n$, then the pole and zero will have minimal effect on the step response.

Suppose that we have

$$T(s) = \frac{1.6(s + 2.5)}{(s^2 + 6s + 25)(0.16s + 1)}.$$

Note that the DC gain is $T(0) = 1$, and we expect a zero steady-state error for a step input. Comparing the two transfer functions, we determine that $\zeta\omega_n = 3$, $\tau = 0.16$, and $a = 2.5$. The poles and the zero are shown on the s-plane in Figure 5.16. As an approximation, we neglect the real pole and zero to obtain

$$T(s) \approx \frac{25}{s^2 + 6s + 25}.$$

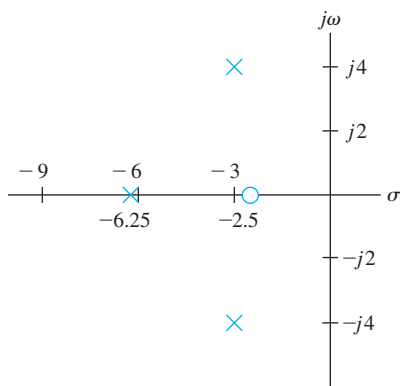


FIGURE 5.16
The poles and zeros
on the s-plane for a
third-order system.

We now have $\zeta = 0.6$ and $\omega_n = 5$ for dominant poles. For this second-order system we expect

$$T_s = \frac{4}{\zeta \omega_n} = 1.33 \text{ s} \quad \text{and} \quad P.O. = 100 e^{-\pi \zeta / \sqrt{1-\zeta^2}} = 9.5\%.$$

For the actual third-order system, we find that the $P.O. = 38\%$ and the $T_s = 1.6$ s. Thus, the effect of the third pole and zero of $T(s)$ cannot be neglected. This is expected since $a \not\gg \zeta \omega_n$ and $\tau \not\ll 1/\zeta \omega_n$. ■

The damping ratio plays a fundamental role in closed-loop system performance. As seen in the design formulas for settling time, percent overshoot, peak time, and rise time, the damping ratio is a key factor in determining the overall performance. In fact, for second-order systems, the damping ratio is the only factor determining the value of the percent overshoot to a step input. As it turns out, the damping ratio can be estimated from the response of a system to a step input [12]. The step response of a second-order system for a unit step input is given in Equation (5.9) yields the frequency of the damped sinusoidal term (for $\zeta < 1$) of

$$\omega = \omega_n (1 - \zeta^2)^{1/2} = \omega_n \beta,$$

and the number of cycles in 1 second is $\omega / (2\pi)$.

The time constant for the exponential decay is $\tau = 1/(\zeta \omega_n)$ in seconds. The number of cycles of the damped sinusoid during one time constant is

$$(\text{cycles/time}) \times \tau = \frac{\omega}{2\pi \zeta \omega_n} = \frac{\omega_n \beta}{2\pi \zeta \omega_n} = \frac{\beta}{2\pi \zeta}.$$

Assuming that the response decays in n visible time constants, we have

$$\text{cycles visible} = \frac{n\beta}{2\pi \zeta}. \quad (5.19)$$

For the second-order system, the response remains within 2% of the steady-state value after four time constants (4τ). Hence, $n = 4$, and

$$\text{cycles visible} = \frac{4\beta}{2\pi \zeta} = \frac{4(1 - \zeta^2)^{1/2}}{2\pi \zeta} \simeq \frac{0.6}{\zeta} \quad (5.20)$$

for $0.2 \leq \zeta \leq 0.6$. From the step response, you count the number of cycles visible up to the settling time, and use Equation (5.20) to estimate ζ ,

An alternative method of estimating ζ is to determine the percent overshoot for the step response and use Equation (5.16) to estimate ζ .

5.5 THE s-PLANE ROOT LOCATION AND THE TRANSIENT RESPONSE

The transient response of a closed-loop feedback control system can be described in terms of the location of the poles of the transfer function. The closed-loop transfer function is written in general as

$$T(s) = \frac{Y(s)}{R(s)} = \frac{\sum P_i(s) \Delta_i(s)}{\Delta(s)},$$

where $\Delta(s) = 0$ is the characteristic equation of the system. For a unity negative feedback control system the characteristic equation reduces to $1 + G_c(s)G(s) = 0$. It is the poles and zeros of $T(s)$ that determine the transient response. However, for a closed-loop system, the poles of $T(s)$ are the roots of the characteristic equation $\Delta(s) = 0$. The output of a system (with DC gain = 1) without repeated roots and a unit step input can be formulated as a partial fraction expansion as

$$Y(s) = \frac{1}{s} + \sum_{i=1}^M \frac{A_i}{s + \sigma_i} + \sum_{k=1}^N \frac{B_k s + C_k}{s^2 + 2\alpha_k s + (\alpha_k^2 + \omega_k^2)}, \quad (5.21)$$

where the A_i , B_k , and C_k are constants. The roots of the system must be either $s = -\sigma_i$ or complex conjugate pairs such as $s = -\alpha_k \pm j\omega_k$. Then the inverse transform results in the transient response as the sum of terms

$$y(t) = 1 + \sum_{i=1}^M A_i e^{-\sigma_i t} + \sum_{k=1}^N D_k e^{-\alpha_k t} \sin(\omega_k t + \theta_k), \quad (5.22)$$

where D_k is a constant and depends on B_k , C_k , α_k , and ω_k . The transient response is composed of the steady-state output, exponential terms, and damped sinusoidal terms. For the response to be stable—that is, bounded for a step input—the real part of the poles must be in the left-hand portion of the s -plane. The impulse response for various root locations is shown in Figure 5.17. The information imparted by the location of the roots is very descriptive.

It is important for the control system designer to understand the complete relationship of the frequency domain representation of a linear system, the poles and zeros of its transfer function, and its time-domain response to step and other inputs. In such areas as signal processing and control, many of the analysis and design calculations are done in the s -plane, where a system model is represented in terms of the poles and zeros of its transfer function $T(s)$. On the other hand, system performance is often analyzed by examining time-domain responses, particularly when dealing with control systems.

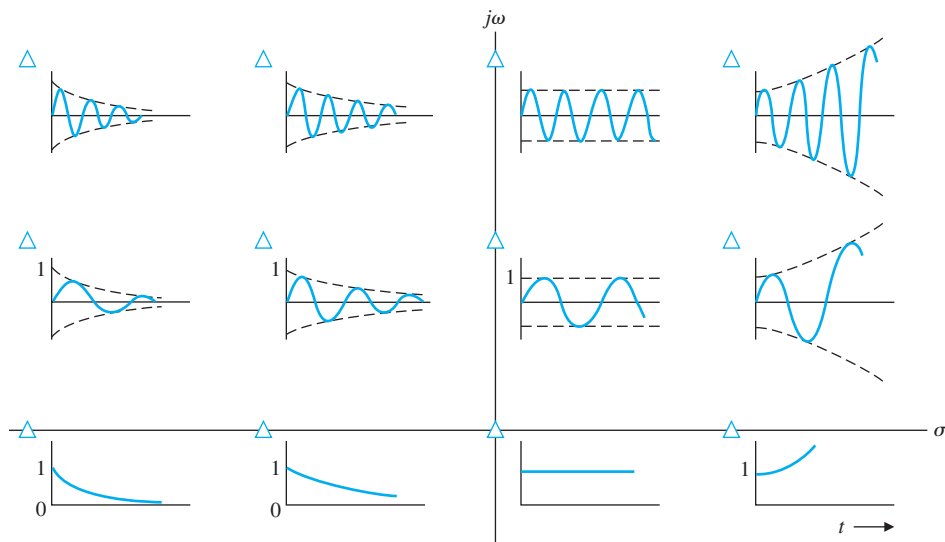


FIGURE 5.17
Impulse response
for various root
locations in the
 s -plane. (The
conjugate root is
not shown.)

The control system designer will envision the effects on the step and impulse responses of adding, deleting, or moving poles and zeros of $T(s)$ in the s -plane. Likewise, the designer should visualize the necessary changes for the poles and zeros of $T(s)$, in order to effect desired changes in the step and impulse responses.

An experienced designer is aware of the effects of zero locations on system response, as well. The poles of $T(s)$ determine the particular response modes that will be present, and the zeros of $T(s)$ establish the relative weightings of the individual mode functions. For example, moving a zero closer to a specific pole will reduce the relative contribution to the output response. In other words, the zeros have a direct impact on the values of A_i and D_k in Equation (5.22). For example, if there is a zero near the pole at $s = -\sigma_i$, then A_i will be much smaller in magnitude.

5.6 THE STEADY-STATE ERROR OF FEEDBACK CONTROL SYSTEMS

One of the fundamental reasons for using feedback, despite its cost and increased complexity, is the attendant improvement in the reduction of the steady-state error of the system. The steady-state error of a stable closed-loop system is usually several orders of magnitude smaller than the error of an open-loop system. The system actuating signal, which is a measure of the system error, is denoted as $E_a(s)$. Consider a unity negative feedback system. In the absence of external disturbances, $T_d(s) = 0$, and measurement noise, $N(s) = 0$, the tracking error is

$$E(s) = \frac{1}{1 + G_c(s)G(s)} R(s).$$

Using the final value theorem and computing the steady-state tracking error yields

$$\lim_{t \rightarrow \infty} e(t) = e_{ss} = \lim_{s \rightarrow 0} s \frac{1}{1 + G_c(s)G(s)} R(s). \quad (5.23)$$

It is useful to determine the steady-state error of the system for the three standard test inputs for the unity feedback system. Later in this section we will consider steady-state tracking errors for nonunity feedback systems.

Step Input. The steady-state error for a step input of magnitude A is therefore

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(A/s)}{1 + G_c(s)G(s)} = \frac{A}{1 + \lim_{s \rightarrow 0} G_c(s)G(s)}.$$

It is the form of the loop transfer function $G_c(s)G(s)$ that determines the steady-state error. The loop transfer function is written in general form as

$$G_c(s)G(s) = \frac{K \prod_{i=1}^M (s + z_i)}{s^N \prod_{k=1}^Q (s + p_k)}, \quad (5.24)$$

where $\prod$ denotes the product of the factors and $z_i \neq 0$, $p_k \neq 0$ for any $1 \leq i \leq M$ and $i \leq k \leq Q$. Therefore, the loop transfer function as s approaches zero depends

on the number of integrations, N . If N is greater than zero, then $\lim_{s \rightarrow 0} G_c(s)G(s)$ approaches infinity, and the steady-state error approaches zero. The number of integrations is often indicated by labeling a system with a **type number** that is equal to N .

Consequently, for a type-zero system, $N = 0$, the steady-state error is

$$e_{ss} = \frac{A}{1 + G_c(0)G(0)} = \frac{A}{1 + K \prod_{i=1}^M z_i / \prod_{k=1}^Q p_k}. \quad (5.25)$$

The constant $G_c(0)G(0)$ is denoted by K_p , the **position error constant**, and is given by

$$K_p = \lim_{s \rightarrow 0} G_c(s)G(s).$$

The steady-state tracking error for a step input of magnitude A is thus given by

$$e_{ss} = \frac{A}{1 + K_p}. \quad (5.26)$$

Hence, the steady-state error for a unit step input with one integration or more, $N \geq 1$, is zero because

$$e_{ss} = \lim_{s \rightarrow 0} \frac{A}{1 + K \prod z_i / (s^N \prod p_k)} = \lim_{s \rightarrow 0} \frac{As^N}{s^N + K \prod z_i / \prod p_k} = 0. \quad (5.27)$$

Ramp Input. The steady-state error for a ramp (velocity) input with a slope A is

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(A/s^2)}{1 + G_c(s)G(s)} = \lim_{s \rightarrow 0} \frac{A}{s + sG_c(s)G(s)} = \lim_{s \rightarrow 0} \frac{A}{sG_c(s)G(s)}. \quad (5.28)$$

Again, the steady-state error depends upon the number of integrations, N . For a type-zero system, $N = 0$, the steady-state error is infinite. For a type-one system, $N = 1$, the error is

$$e_{ss} = \lim_{s \rightarrow 0} \frac{A}{sK \prod (s + z_i) / [s \prod (s + p_k)]},$$

or

$$e_{ss} = \frac{A}{K \prod z_i / \prod p_k} = \frac{A}{K_v}, \quad (5.29)$$

where K_v is designated the **velocity error constant**. The velocity error constant is computed as

$$K_v = \lim_{s \rightarrow 0} sG_c(s)G(s).$$

When the transfer function possesses two or more integrations, $N \geq 2$, we obtain a steady-state error of zero. When $N = 1$, a steady-state error exists. However,

Table 5.2 Summary of Steady-State Errors

Number of Integrations in $G_c(s)G(s)$, Type Number	Input		
	Step, $r(t) = A$, $R(s) = A/s$	Ramp, $r(t) = At$, $R(s) = A/s^2$	Parabola, $r(t) = At^2/2$, $R(s) = A/s^3$
0	$e_{ss} = \frac{A}{1 + K_p}$	∞	∞
1	$e_{ss} = 0$	$\frac{A}{K_v}$	∞
2	$e_{ss} = 0$	0	$\frac{A}{K_a}$

the steady-state velocity of the output is equal to the velocity of input, as we shall see shortly.

Acceleration Input. When the system input is $r(t) = At^2/2$, the steady-state error is

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(A/s^3)}{1 + G_c(s)G(s)} = \lim_{s \rightarrow 0} \frac{A}{s^2 G_c(s)G(s)}. \quad (5.30)$$

The steady-state error is infinite for one integration. For two integrations, $N = 2$, and we obtain

$$e_{ss} = \frac{A}{K \prod z_i / \prod p_k} = \frac{A}{K_a}, \quad (5.31)$$

where K_a is designated the **acceleration error constant**. The acceleration error constant is

$$K_a = \lim_{s \rightarrow 0} s^2 G_c(s)G(s).$$

When the number of integrations equals or exceeds three, then the steady-state error of the system is zero.

Control systems are often described in terms of their type number and the error constants, K_p , K_v , and K_a . Definitions for the error constants and the steady-state error for the three inputs are summarized in Table 5.2.

EXAMPLE 5.3 Mobile robot steering control

A mobile robot may be designed as an assisting device or servant for a severely disabled person [7]. The steering control system for such a robot can be represented by the block diagram shown in Figure 5.18. The steering controller is

$$G_c(s) = K_1 + K_2/s. \quad (5.32)$$

FIGURE 5.18
Block diagram of steering control system for a mobile robot.

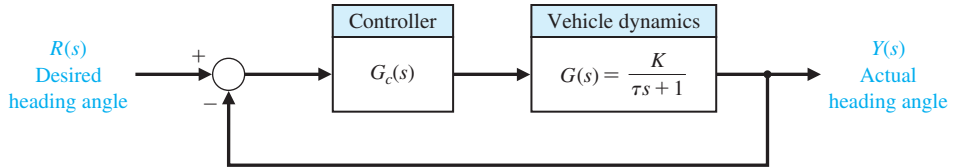
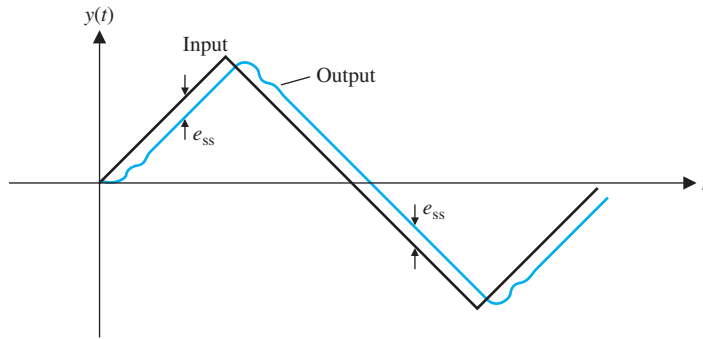


FIGURE 5.19
Triangular wave response.



Therefore, the steady-state error of the system for a step input when $K_2 = 0$ and $G_c(s) = K_1$ is

$$e_{ss} = \frac{A}{1 + K_p}, \quad (5.33)$$

where $K_p = KK_1$. When K_2 is greater than zero, we have a type-1 system,

$$G_c(s) = \frac{K_1s + K_2}{s},$$

and the steady-state error is zero for a step input.

If the steering command is a ramp input, the steady-state error is

$$e_{ss} = \frac{A}{K_v}, \quad (5.34)$$

where

$$K_v = \lim_{s \rightarrow 0} sG_c(s)G(s) = K_2 K.$$

The transient response of the vehicle to a triangular wave input when $G_c(s) = (K_1s + K_2)/s$ is shown in Figure 5.19. The transient response clearly shows the effect of the steady-state error, which may not be objectionable if K_v is sufficiently large. Note that the output attains the desired velocity as required by the input, but it exhibits a steady-state error. ■

The control system error constants, K_p , K_v , and K_a , describe the ability of a system to reduce or eliminate the steady-state error. Therefore, they are utilized as numerical measures of the steady-state performance. The designer determines the error constants for a given system and attempts to determine methods of increasing

the error constants while maintaining an acceptable transient response. In the case of the steering control system, we want to increase the gain factor KK_2 in order to increase K_v and reduce the steady-state error. However, an increase in KK_2 results in an attendant decrease in the system damping ratio ζ and therefore a more oscillatory response to a step input. Thus, we seek a compromise that provides the largest K_v based on the smallest ζ allowable.

In the preceding discussions, we assumed that we had a unity feedback system. Now we consider nonunity feedback systems. For a system in which the feedback is not unity, the units of the output $Y(s)$ are usually different from the output of the sensor. For example, a speed control system is shown in Figure 5.20. The constants K_1 and K_2 account for the conversion of one set of units to another set of units (here we convert rad/s to volts). We can select K_1 , and thus we set $K_1 = K_2$ and move the block for K_1 and K_2 past the summing node. Then we obtain the equivalent block diagram shown in Figure 5.21. Thus, we obtain a unity feedback system as desired.

Consider a nonunity negative feedback system with the system $H(s)$ in the feedback loop given by

$$H(s) = \frac{K_2}{\tau s + 1}$$

which has a DC gain of

$$\lim_{s \rightarrow 0} H(s) = K_2.$$

If we set $K_2 = K_1$, then the system is transformed to that of Figure 5.21 for the steady-state calculation. To see this, consider error of the system $E(s)$, where

$$E(s) = R(s) - Y(s) = [1 - T(s)]R(s), \quad (5.35)$$

since $Y(s) = T(s)R(s)$. Note that

$$T(s) = \frac{K_1 G_c(s)G(s)}{1 + H(s)G_c(s)G(s)} = \frac{(\tau s + 1)K_1 G_c(s)G(s)}{\tau s + 1 + K_1 G_c(s)G(s)},$$

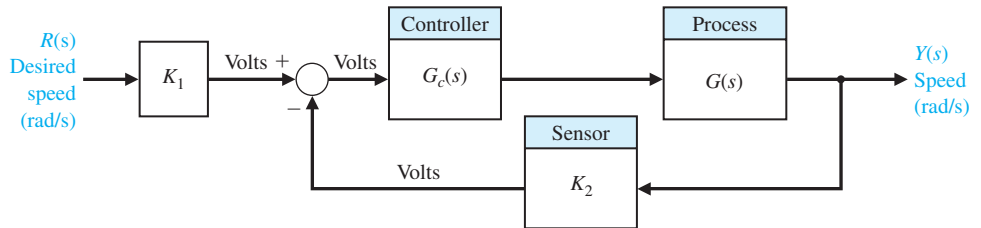


FIGURE 5.20
A speed control system.

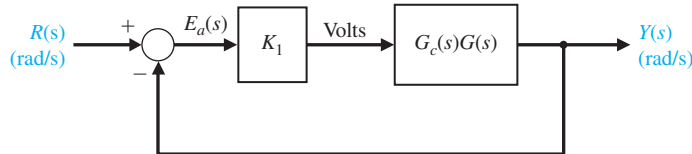


FIGURE 5.21
The speed control system of Figure 5.20 with $K_1 = K_2$.

and therefore,

$$E(s) = \frac{1 + \tau s(1 - K_1 G_c(s)G(s))}{\tau s + 1 + K_1 G_c(s)G(s)} R(s).$$

Then the steady-state error for a unit step input is

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \frac{1}{1 + K_1 \lim_{s \rightarrow 0} G_c(s)G(s)}. \quad (5.36)$$

We assume here that

$$\lim_{s \rightarrow 0} s G_c(s)G(s) = 0.$$

EXAMPLE 5.4 Steady-state error

Let us determine the appropriate value of K_1 and calculate the steady-state error for a unit step input for the system shown in Figure 4.4 when

$$G_c(s) = 40, G(s) = \frac{1}{s + 5}, \text{ and } H(s) = \frac{2}{0.1s + 1}.$$

Selecting $K_1 = K_2 = 2$, we can use Equation (5.36) to determine

$$e_{ss} = \frac{1}{1 + K_1 \lim_{s \rightarrow 0} G_c(s)G(s)} = \frac{1}{1 + 2(40)(1/5)} = \frac{1}{17},$$

or 5.9% of the magnitude of the step input. ■

EXAMPLE 5.5 Nonunity feedback control system

Let us consider the system of Figure 5.22, where we assume we cannot insert a gain K_1 following $R(s)$ as we did for the system of Figure 5.20. Then the actual error is given by Equation (5.35), which is

$$E(s) = [1 - T(s)]R(s).$$

Let us determine an appropriate gain K so that the steady-state error to a step input is minimized. The steady-state error is

$$e_{ss} = \lim_{s \rightarrow 0} s[1 - T(s)]\frac{1}{s},$$

where

$$T(s) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)H(s)} = \frac{K(s + 4)}{(s + 2)(s + 4) + 2K}.$$

Then we have

$$T(0) = \frac{4K}{8 + 2K}.$$

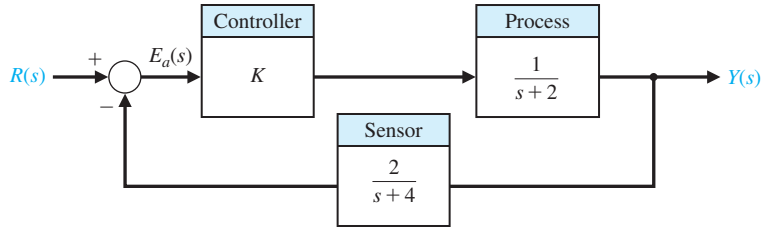


FIGURE 5.22
A system with a feedback $H(s)$.

The steady-state error for a unit step input is

$$e_{ss} = 1 - T(0).$$

Thus, to achieve a zero steady-state error, we require that

$$T(0) = \frac{4K}{8 + 2K} = 1,$$

or $8 + 2K = 4K$. Thus, $K = 4$ will yield a zero steady-state error. It is unlikely that meeting a steady-state error specification is the only requirement of the feedback control system, so choosing the control as a gain with only one parameter to adjust is probably not practical. ■

The determination of the steady-state error is simpler for unity feedback systems. However, it is possible to extend the notion of error constants to nonunity feedback systems by first appropriately rearranging the block diagram to obtain an equivalent unity feedback system. Remember that the underlying system must be stable, otherwise our use of the final value theorem will be compromised. Consider the nonunity feedback system in Figure 5.21 and assume that $K_1 = 1$. The closed-loop transfer function is

$$\frac{Y(s)}{R(s)} = T(s) = \frac{G_c(s)G(s)}{1 + H(s)G_c(s)G(s)}.$$

By manipulating the block diagram appropriately we can obtain the equivalent unity feedback system with

$$\frac{Y(s)}{R(s)} = T(s) = \frac{Z(s)}{1 + Z(s)} \quad \text{where} \quad Z(s) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)(H(s) - 1)}.$$

The loop transfer function of the equivalent unity feedback system is $Z(s)$. It follows that the error constants for nonunity feedback systems are given as:

$$K_p = \lim_{s \rightarrow 0} Z(s), \quad K_v = \lim_{s \rightarrow 0} sZ(s), \quad \text{and} \quad K_a = \lim_{s \rightarrow 0} s^2 Z(s).$$

Note that when $H(s) = 1$, then $Z(s) = G_c(s)G(s)$ and we maintain the unity feedback error constants. For example, when $H(s) = 1$, then $K_p = \lim_{s \rightarrow 0} Z(s) = \lim_{s \rightarrow 0} G_c(s)G(s)$, as expected.

5.7 PERFORMANCE INDICES

Modern control theory assumes that we can specify quantitatively the required system performance. Then a performance index can be calculated or measured and used to evaluate the system performance. Quantitative measures of the performance of a system are very valuable in the design and operation of control systems.

A system is considered an **optimum control system** when the system parameters are adjusted so that the index reaches an extremum, commonly a minimum value. To be useful, a performance index must be a number that is always positive or zero. Then the best system is defined as the system that minimizes this index.

A performance index is a quantitative measure of the performance of a system and is chosen so that emphasis is given to the important system specifications.

A common performance index is the integral of the square of the error, ISE, which is defined as

$$\text{ISE} = \int_0^T e^2(t) dt. \quad (5.37)$$

The upper limit T is a finite time selected by the control system designer. It is convenient to choose T as the settling time T_s . The step response for a specific feedback control system is shown in Figure 5.23(b), and the error in Figure 5.23(c). The error squared is shown in Figure 5.23(d), and the integral of the error squared in Figure 5.23(e). This criterion will discriminate between excessively overdamped and excessively underdamped systems. The minimum value of the integral occurs for a compromise value of the damping. The performance index of Equation (5.37) is mathematically convenient for analytical and computational purposes.

Three other performance indices we might consider include

$$\text{IAE} = \int_0^T |e(t)| dt, \quad (5.38)$$

$$\text{ITAE} = \int_0^T t |e(t)| dt, \quad (5.39)$$

and

$$\text{ITSE} = \int_0^T te^2(t) dt. \quad (5.40)$$

The ITAE is able to reduce the contribution of any large initial errors, as well as to emphasize errors occurring later in the response [6]. The performance index ITAE provides the best selectivity of the performance indices; that is, the minimum value of the integral is readily discernible as the system parameters are varied.

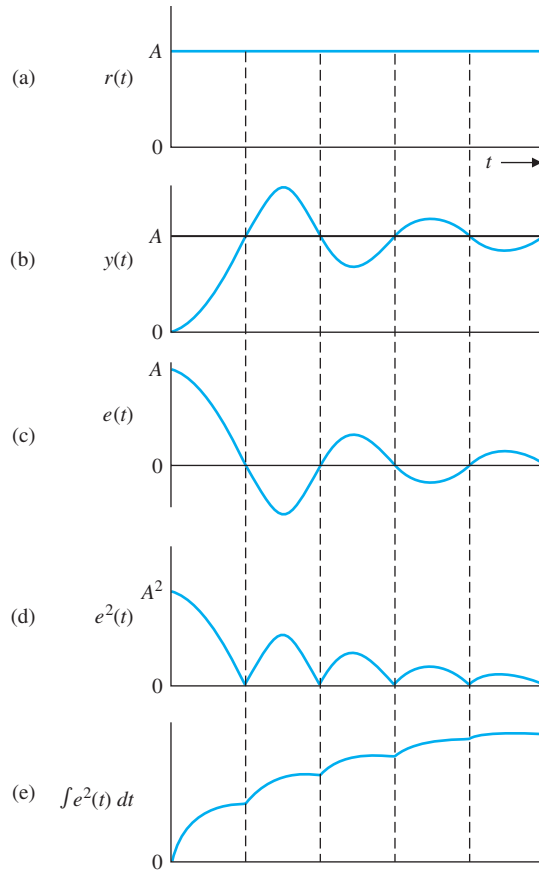


FIGURE 5.23
The calculation
of the Integral
squared error.

The general form of the performance integral is

$$I = \int_0^T f(e(t), r(t), y(t), t) dt, \quad (5.41)$$

where f is a function of the error, input, output, and time. We can obtain numerous indices based on various combinations of the system variables and time.

EXAMPLE 5.6 Space telescope control system

Consider a space telescope pointing control system shown in Figure 5.24 [9]. We desire to select the magnitude of the gain, K_3 , to minimize the effect of the disturbance, $T_d(s)$. The closed-loop transfer function from the disturbance to the output is

$$\frac{Y(s)}{T_d(s)} = \frac{s(s + K_1 K_3)}{s^2 + K_1 K_3 s + K_1 K_2 K_p}. \quad (5.42)$$

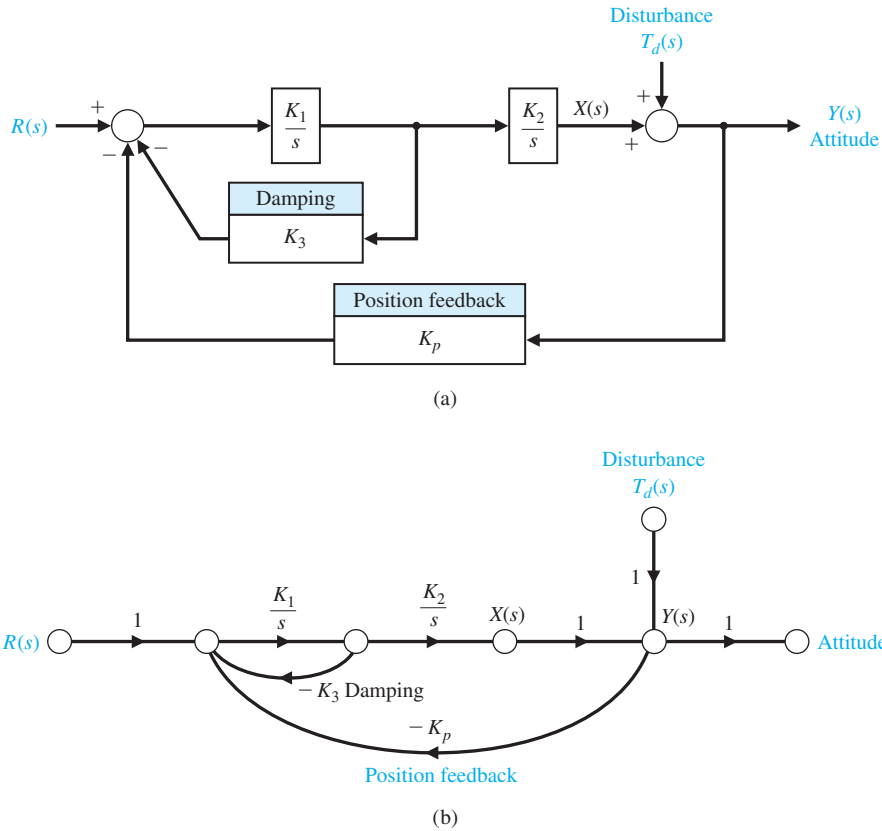


FIGURE 5.24
A space telescope pointing control system. (a) Block diagram. (b) Signal-flow graph.

Typical values for the constants are $K_1 = 0.5$ and $K_1 K_2 K_p = 2.5$. In this case, the goal is to minimize $y(t)$, where, for a unit step disturbance, the minimum ISE can be analytically calculated. The attitude is

$$y(t) = \frac{\sqrt{10}}{\beta} \left[e^{-0.25 K_3 t} \sin \left(\frac{\beta}{2} t + \psi \right) \right], \quad (5.43)$$

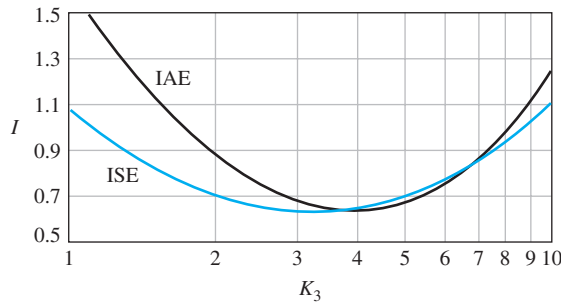
where $\beta = \sqrt{10 - K_3^2/4}$. Squaring $y(t)$ and integrating the result yields

$$\begin{aligned} I &= \int_0^\infty \frac{10}{\beta^2} e^{-0.5 K_3 t} \sin^2 \left(\frac{\beta}{2} t + \psi \right) dt = \int_0^\infty \frac{10}{\beta^2} e^{-0.5 K_3 t} \left(\frac{1}{2} - \frac{1}{2} \cos(\beta t + 2\psi) \right) dt \\ &= \frac{1}{K_3} + 0.1 K_3. \end{aligned} \quad (5.44)$$

Differentiating I and equating the result to zero, and solving for K_3 , we obtain

$$\frac{dI}{dK_3} = -K_3^{-2} + 0.1 = 0. \quad (5.45)$$

FIGURE 5.25
The performance indices of the telescope control system versus K_3 .



Therefore, the minimum ISE is obtained when $K_3 = \sqrt{10} = 3.2$. This value of K_3 corresponds to a damping ratio $\zeta = 0.50$. The values of ISE and IAE for this system are plotted in Figure 5.25. The minimum for the IAE performance index is obtained when $K_3 = 4.2$ and $\zeta = 0.665$. While the ISE criterion is not as selective as the IAE criterion, it is clear that it is possible to solve analytically for the minimum value of ISE. The minimum of IAE is obtained by computing the actual value of IAE for several values of the parameter of interest. ■

A control system is optimum when the selected performance index is minimized. However, the optimum value of the parameters depends directly on the definition of optimum, that is, the performance index. Therefore, in Example 5.6, we found that the optimum setting varied for different performance indices.

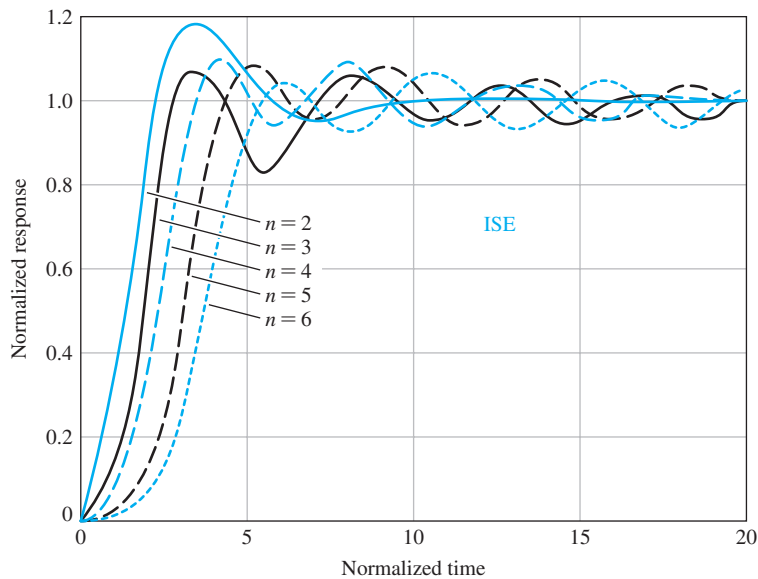
The coefficients that will minimize the ITAE performance criterion for a step input have been determined for the general closed-loop transfer function [6]

$$T(s) = \frac{Y(s)}{R(s)} = \frac{b_0}{s^n + b_{n-1}s^{n-1} + \cdots + b_1s + b_0}. \quad (5.46)$$

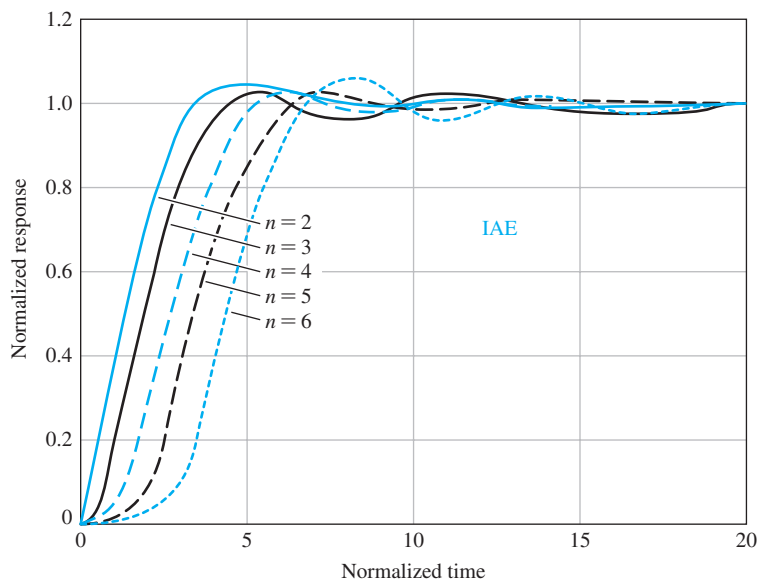
This transfer function has a steady-state error equal to zero for a step input. Note that the transfer function has n poles and no zeros. The optimum coefficients for the ITAE criterion are given in Table 5.3. The responses using optimum coefficients for a step input are given in Figure 5.26 for ISE, IAE, and ITAE. The responses are provided for normalized time $\omega_n t$. Other standard forms based on different performance indices are available and can be useful in aiding the designer to determine the range of coefficients for a specific problem.

Table 5.3 The Optimum Coefficients of $T(s)$ Based on the ITAE Criterion for a Step Input

$s + \omega_n$
$s^2 + 1.4\omega_n s + \omega_n^2$
$s^3 + 1.75\omega_n s^2 + 2.15\omega_n^2 s + \omega_n^3$
$s^4 + 2.1\omega_n s^3 + 3.4\omega_n^2 s^2 + 2.7\omega_n^3 s + \omega_n^4$
$s^5 + 2.8\omega_n s^4 + 5.0\omega_n^2 s^3 + 5.5\omega_n^3 s^2 + 3.4\omega_n^4 s + \omega_n^5$
$s^6 + 3.25\omega_n s^5 + 6.60\omega_n^2 s^4 + 8.60\omega_n^3 s^3 + 7.45\omega_n^4 s^2 + 3.95\omega_n^5 s + \omega_n^6$



(a)



(b)

FIGURE 5.26 Step responses of a normalized transfer function using optimum coefficients for (a) ISE, (b) IAE, and (c) ITAE. The response is for normalized time, $\omega_n t$.

For a ramp input, the coefficients have been determined that minimize the ITAE criterion for the general closed-loop transfer function [6]

$$T(s) = \frac{b_1 s + b_0}{s^n + b_{n-1} s^{n-1} + \cdots + b_1 s + b_0}. \quad (5.47)$$

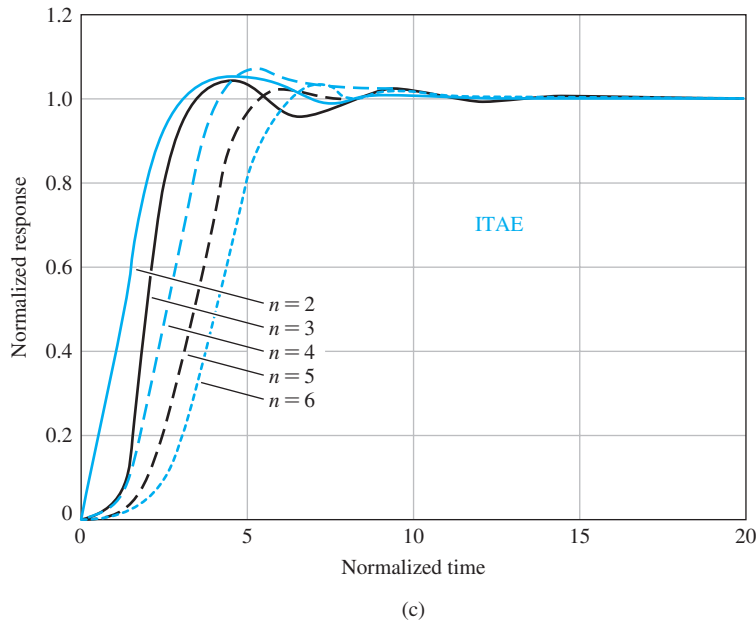


FIGURE 5.26
(Continued)

Table 5.4 The Optimum Coefficients of $T(s)$ Based on the ITAE Criterion for a Ramp Input

$$\begin{array}{l}
 s^2 + 3.2\omega_n s + \omega_n^2 \\
 s^3 + 1.75\omega_n s^2 + 3.25\omega_n^2 s + \omega_n^3 \\
 s^4 + 2.41\omega_n s^3 + 4.93\omega_n^2 s^2 + 5.14\omega_n^3 s + \omega_n^4 \\
 s^5 + 2.19\omega_n s^4 + 6.50\omega_n^2 s^3 + 6.30\omega_n^3 s^2 + 5.24\omega_n^4 s + \omega_n^5
 \end{array}$$

This transfer function has a steady-state error equal to zero for a ramp input. The optimum coefficients for this transfer function are given in Table 5.4. The transfer function, Equation (5.50), implies that the process $G(s)$ has two or more pure integrations, as required to provide zero steady-state error.

5.8 THE SIMPLIFICATION OF LINEAR SYSTEMS

It is quite useful to study complex systems with high-order transfer functions by using lower-order approximate models. Several methods are available for reducing the order of a systems transfer function. One relatively simple way to delete a certain insignificant pole of a transfer function is to note a pole that has a negative real part that is much more negative than the other poles. Thus, that pole is expected to affect the transient response insignificantly.

For example, if we have a system with transfer function

$$G(s) = \frac{K}{s(s+2)(s+30)},$$

we can safely neglect the impact of the pole at $s = -30$. However, we must retain the steady-state response of the system, so we reduce the system to

$$G(s) = \frac{(K/30)}{s(s+2)}.$$

A more sophisticated approach attempts to match the frequency response of the reduced-order transfer function with the original transfer function frequency response as closely as possible. Although frequency response methods are covered in Chapter 8, the associated approximation method strictly relies on algebraic manipulation and is presented here. Consider the high-order system be described by the transfer function

$$G_H(s) = K \frac{a_m s^m + a_{m-1} s^{m-1} + \cdots + a_1 s + 1}{b_n s^n + b_{n-1} s^{n-1} + \cdots + b_1 s + 1}, \quad (5.48)$$

in which the poles are in the left-hand s -plane and $m \leq n$. The lower-order approximate transfer function is

$$G_L(s) = K \frac{c_p s^p + \cdots + c_1 s + 1}{d_g s^g + \cdots + d_1 s + 1}, \quad (5.49)$$

where $p \leq g < n$. Notice that the gain constant, K , is the same for the original and approximate system; this ensures the same steady-state response. The method outlined in Example 5.7 is based on selecting c_i and d_i in such a way that $G_L(s)$ has a frequency response very close to that of $G_H(s)$. This is equivalent to stating that $G_H(j\omega)/G_L(j\omega)$ is required to deviate the least amount from unity for various frequencies. The c and d coefficients are obtained via

$$M^{(k)}(s) = \frac{d^k}{ds^k} M(s) \quad (5.50)$$

and

$$\Delta^{(k)}(s) = \frac{d^k}{ds^k} \Delta(s), \quad (5.51)$$

where $M(s)$ and $\Delta(s)$ are the numerator and denominator polynomials of $G_H(s)/G_L(s)$, respectively. We also define

$$M_{2q} = \sum_{k=0}^{2q} \frac{(-1)^{k+q} M^{(k)}(0) M^{(2q-k)}(0)}{k! (2q-k)!}, \quad q = 0, 1, 2, \dots \quad (5.52)$$

and an analogous equation for Δ_{2q} . The solutions for the c and d coefficients are obtained by equating

$$M_{2q} = \Delta_{2q} \quad (5.53)$$

for $q = 1, 2, \dots$ up to the number required to solve for the unknown coefficients.

EXAMPLE 5.7 A simplified model

Consider the third-order system

$$G_H(s) = \frac{6}{s^3 + 6s^2 + 11s + 6} = \frac{1}{1 + \frac{11}{6}s + s^2 + \frac{1}{6}s^3}. \quad (5.54)$$

Using the second-order model

$$G_L(s) = \frac{1}{1 + d_1s + d_2s^2}, \quad (5.55)$$

we determine that

$$M(s) = 1 + d_1s + d_2s^2, \quad \text{and} \quad \Delta(s) = 1 + \frac{11}{6}s + s^2 + \frac{1}{6}s^3.$$

Then we know that

$$M^{(0)}(s) = 1 + d_1s + d_2s^2, \quad (5.56)$$

and $M^{(0)}(0) = 1$. Similarly, we have

$$M^{(1)} = \frac{d}{ds}(1 + d_1s + d_2s^2) = d_1 + 2d_2s. \quad (5.57)$$

Therefore, $M^{(1)}(0) = d_1$. Continuing this process, we find that

$$\begin{aligned} M^{(0)}(0) &= 1 & \Delta^{(0)}(0) &= 1, \\ M^{(1)}(0) &= d_1 & \Delta^{(1)}(0) &= \frac{11}{6}, \\ M^{(2)}(0) &= 2d_2 & \Delta^{(2)}(0) &= 2, \\ M^{(3)}(0) &= 0 & \Delta^{(3)}(0) &= 1. \end{aligned} \quad (5.58)$$

We now equate $M_{2q} = \Delta_{2q}$ for $q = 1$ and 2 . We find that, for $q = 1$,

$$\begin{aligned} M_2 &= (-1) \frac{M^{(0)}(0)M^{(2)}(0)}{2} + \frac{M^{(1)}(0)M^{(1)}(0)}{1} + (-1) \frac{M^{(2)}(0)M^{(0)}(0)}{2} \\ &= -d_2 + d_1^2 - d_2 = -2d_2 + d_1^2. \end{aligned} \quad (5.59)$$

Since the equation for Δ_2 is similar, we have

$$\begin{aligned} \Delta_2 &= (-1) \frac{\Delta^{(0)}(0)\Delta^{(2)}(0)}{2} + \frac{\Delta^{(1)}(0)\Delta^{(1)}(0)}{1} + (-1) \frac{\Delta^{(2)}(0)\Delta^{(0)}(0)}{2} \\ &= -1 + \frac{121}{36} - 1 = \frac{49}{36}. \end{aligned} \quad (5.60)$$

Equation (5.53) with $q = 1$ requires that $M_2 = \Delta_2$; therefore,

$$-2d_2 + d_1^2 = \frac{49}{36}. \quad (5.61)$$

Completing the process for $M_4 = \Delta_4$, we obtain

$$d_2^2 = \frac{7}{18}. \quad (5.62)$$

Solving Equations (5.61) and (5.62) yields $d_1 = 1.615$ and $d_2 = 0.624$. (The other sets of solutions are rejected because they lead to unstable poles.) The lower-order system transfer function is

$$G_L(s) = \frac{1}{1 + 1.615s + 0.624s^2} = \frac{1.60}{s^2 + 2.590s + 1.60}. \quad (5.63)$$

It is interesting to see that the poles of $G_H(s)$ are $s = -1, -2, -3$, whereas the poles of $G_L(s)$ are $s = -1.024$ and -1.565 . Because the lower-order model has two poles, we estimate that we would obtain a slightly overdamped step response with a settling time to within 2% of the final value in approximately 3 seconds. ■

It is sometimes desirable to retain the dominant poles of the original system, $G_H(s)$, in the low-order model. This can be accomplished by specifying the denominator of $G_L(s)$ to be the dominant poles of $G_H(s)$ and allowing the numerator of $G_L(s)$ to be subject to approximation.

Another novel and useful method for reducing the order is the Routh approximation method based on the idea of truncating the Routh table used to determine stability. The Routh approximants can be computed by a finite recursive algorithm [19].

5.9 DESIGN EXAMPLES

In this section we present two illustrative examples. The first example is a simplified view of the Hubble space telescope pointing control problem. The Hubble space telescope problem highlights the process of computing controller gains to achieve desired percent overshoot specifications, as well as meeting steady-state error specifications. The second example considers the control of the bank angle of an airplane. The airplane attitude motion control example represents a more in-depth look at the control design problem. Here we consider a complex fourth-order model of the lateral dynamics of the aircraft motion that is approximated by a second-order model using the approximation methods of Section 5.8. The simplified model can be used to gain insight into the controller design and the impact of key controller parameters on the transient performance.

EXAMPLE 5.8 Hubble space telescope control

The orbiting Hubble space telescope is the most complex and expensive scientific instrument that has ever been built. The telescope's 2.4 meter mirror has the

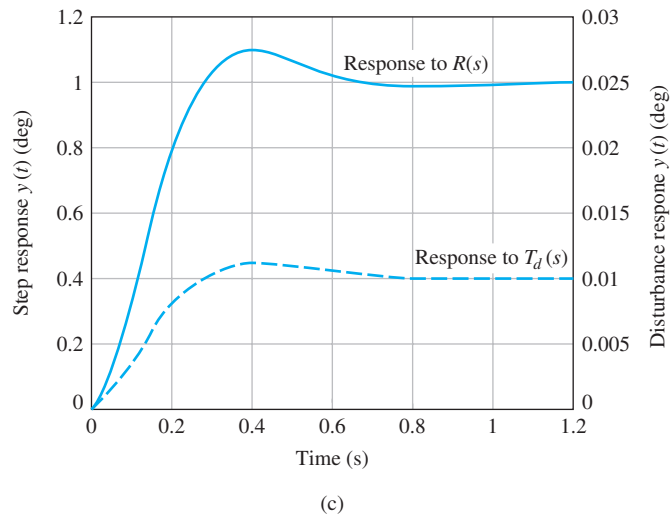
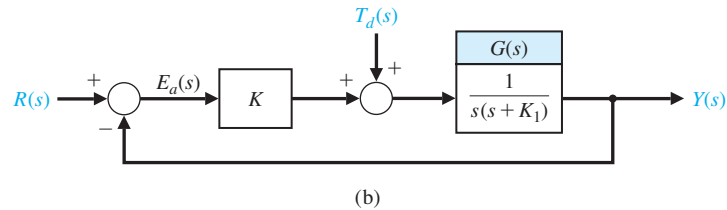
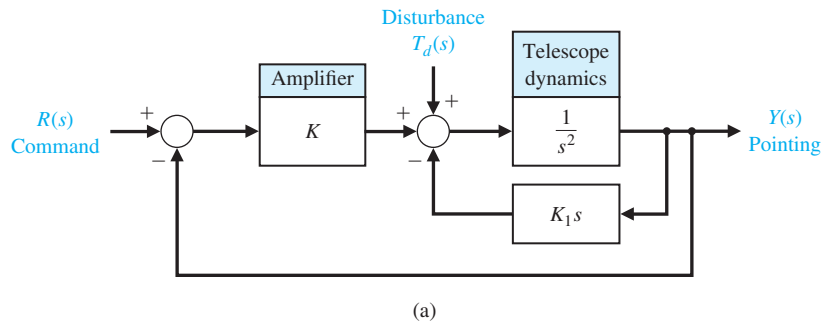


FIGURE 5.27
 (a) The Hubble telescope pointing system, (b) reduced block diagram, and (c) system response to a unit step input command and a unit step disturbance input.

smoothest surface of any mirror made, and its pointing system can center it on a dime 400 miles away [18, 21]. Consider the model of the telescope-pointing system shown in Figure 5.27.

The goal of the design is to choose K_1 and K so that (1) the percent overshoot of the output to a step command, $r(t)$, is $P.O. \leq 10\%$, (2) the steady-state error to a ramp command is minimized, and (3) the effect of a step disturbance is reduced. Since the system has an inner loop, block diagram reduction can be used to obtain the simplified system of Figure 5.27(b).

The output due to the two inputs of the system of Figure 5.27(b) is given by

$$Y(s) = T(s)R(s) + [T(s)/K]T_d(s), \quad (5.64)$$

where

$$T(s) = \frac{KG(s)}{1 + KG(s)} = \frac{L(s)}{1 + L(s)}.$$

The tracking error is

$$E(s) = \frac{1}{1 + L(s)}R(s) - \frac{G(s)}{1 + L(s)}T_d(s). \quad (5.65)$$

First, let us select K and K_1 to meet the percent overshoot requirement for a step input, $R(s) = A/s$. Setting $T_d(s) = 0$, we have

$$Y(s) = \frac{KG(s)}{1 + KG(s)}R(s) = \frac{K}{s^2 + K_1s + K} \left(\frac{A}{s} \right). \quad (5.66)$$

To set the percent overshoot to $P.O. \leq 10\%$, we select $\zeta = 0.6$. We can use Equation (5.16) to determine that $P.O. = 9.5\%$ for $\zeta = 0.6$. We next examine the steady-state error for a ramp, $r(t) = Bt, t \geq 0$. Using Equation 5.28 we find

$$e_{ss} = \lim_{s \rightarrow 0} \left\{ \frac{B}{sKG(s)} \right\} = \frac{B}{K/K_1}. \quad (5.67)$$

The steady-state error due to the ramp disturbance is reduced by increasing KK_1 . The steady-state error due to a unit step disturbance is equal to $-1/K$. The steady-state error due to the step disturbance input can thus be reduced by increasing K . In summary, we seek a large K and a large value of K/K_1 to obtain low steady-state errors due to a step and ramp disturbance, respectively. We also require $\zeta = 0.6$ to limit the percent overshoot.

With $\zeta = 0.6$, the characteristic equation of the system is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 2(0.6)\omega_n s + K. \quad (5.68)$$

Therefore, $\omega_n = \sqrt{K}$, and the second term of the denominator of Equation (5.69) requires $K_1 = 2(0.6)\omega_n$. Then $K_1 = 1.2\sqrt{K}$, so the ratio K/K_1 is

$$\frac{K}{K_1} = \frac{K}{1.2\sqrt{K}} = \frac{\sqrt{K}}{1.2}.$$

If we select $K = 100$, we have $K_1 = 12$ and $K/K_1 = 8.33$. The responses of the system to a unit step input command and a unit step disturbance input are shown in Figure 5.27(c). Note how the effect of the disturbance is relatively insignificant. Finally, we note that the steady-state error for a ramp input is

$$e_{ss} = \frac{B}{8.33} = 0.12B.$$

This design, using $K = 100$, provides acceptable results. ■

EXAMPLE 5.9 Attitude control of an airplane

Each time we fly on a commercial airliner, we experience first-hand the benefits of automatic control systems. These systems assist pilots by improving the handling qualities of the aircraft over a wide range of flight conditions and by providing pilot relief (for such emergencies as going to the restroom) during extended flights. The special relationship between flight and controls began in the early work of the Wright brothers. Using wind tunnels, the Wright brothers applied systematic design techniques to make their dream of powered flight a reality. This systematic approach to design contributed to their success.

Another significant aspect of their approach was their emphasis on flight controls; the brothers insisted that their aircraft be pilot-controlled. Observing birds control their rolling motion by twisting their wings, the Wright brothers built aircraft with mechanical mechanisms that twisted their airplane wings. Today we no longer use wing warping as a mechanism for performing a roll maneuver; instead we control rolling motion by using ailerons, as shown in Figure 5.28. The Wright brothers also used elevators (located forward) for longitudinal control (pitch motion) and rudders for lateral control (yaw motion). Today's aircraft still use both elevators and rudders, although the elevators are generally located on the tail (rearward).

The first controlled, powered, unassisted take-off flight occurred in 1903 with the *Wright Flyer I* (a.k.a. *Kitty Hawk*). The first practical airplane, the *Flyer III*, could fly figure eights and stay aloft for half an hour. Three-axis flight control was a major (and often overlooked) contribution of the Wright brothers. A concise historical perspective is presented in Stevens and Lewis [24]. The continuing desire to fly faster, lighter, and longer fostered further developments in automatic flight control.

The main topic of this chapter is control of the automatic rolling motion of an airplane. The elements of the design process emphasized in this chapter are illustrated in Figure 5.29.

We begin by considering the model of the lateral dynamics of an airplane moving along a steady, wings-level flight path. By lateral dynamics, we mean the attitude motion of the aircraft about the forward velocity. An accurate mathematical model describing the motion (translational and rotational) of an aircraft is a complicated set of highly nonlinear, time-varying, coupled differential equations. A good description of the process of developing such a mathematical model appears in Etkin and Reid [25].

For our purposes a simplified dynamic model is required for the autopilot design process. A simplified model might consist of a transfer function describing

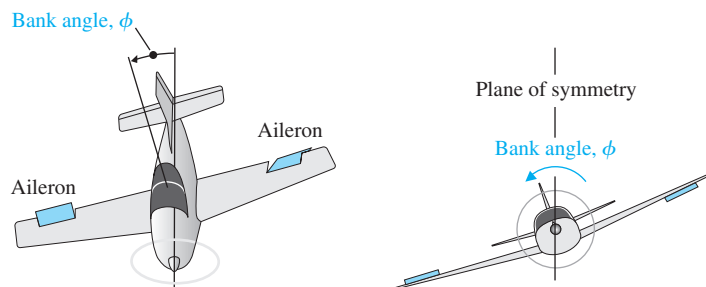


FIGURE 5.28
Control of the bank angle of an airplane using differential deflections of the ailerons.

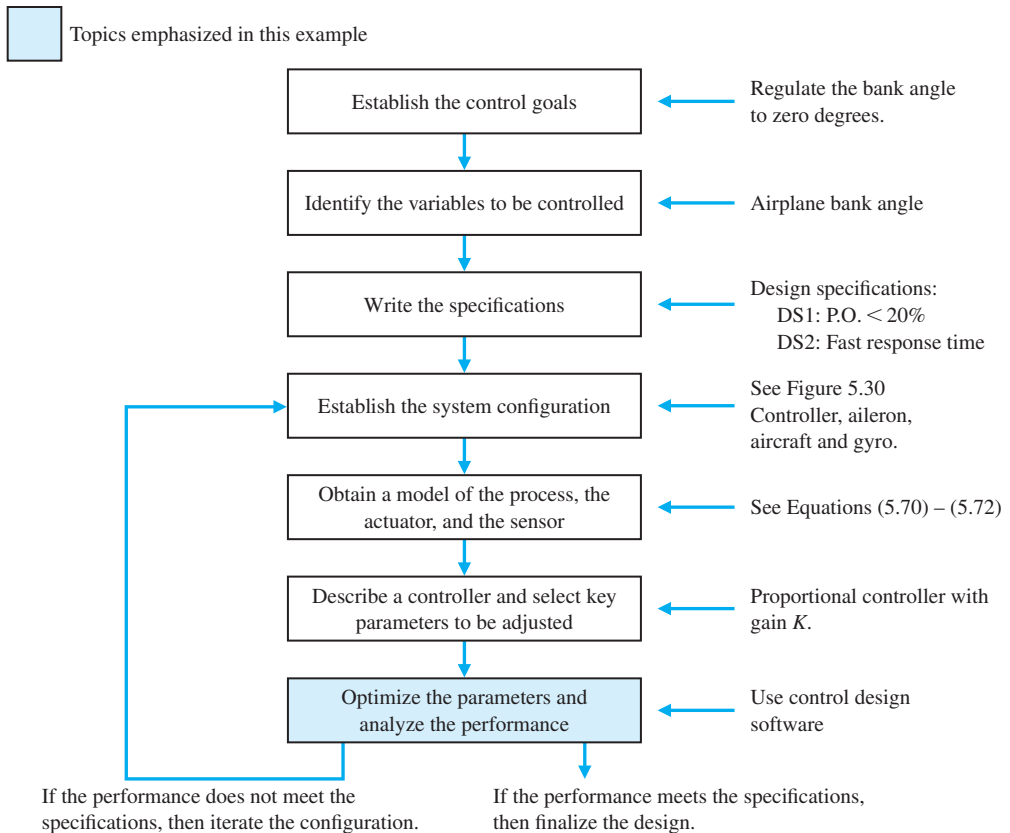


FIGURE 5.29 Elements of the control system design process emphasized in the airplane attitude control example.

the input–output relationship between the aileron deflection and the aircraft bank angle. Obtaining such a transfer function would require many prudent simplifications to the original high-fidelity, nonlinear mathematical model.

Suppose we have a rigid aircraft with a plane of symmetry. The airplane is assumed to be cruising at subsonic or low supersonic (Mach Equations <3) speeds. This allows us to make a flat-Earth approximation. We ignore any rotor gyroscopic effects due to spinning masses on the aircraft (such as propellers or turbines). These assumptions allow us to decouple the longitudinal rotational (pitching) motion from the lateral rotational (rolling and yawing) motion.

Of course, we also need to consider a linearization of the nonlinear equations of motion. To accomplish this, we consider only steady-state flight conditions such as

- ☐ Steady, wings-level flight
- ☐ Steady, level turning flight
- ☐ Steady, symmetric pull-up
- ☐ Steady roll.

For this example we assume that the airplane is flying at low speed in a steady, wings-level attitude, and we want to design an autopilot to control the rolling motion. We can state the control goal as follows:

Control Goal

Regulate the airplane bank angle to zero degrees (steady, wings level) and maintain the wings-level orientation in the presence of unpredictable external disturbances.

We identify the variable to be controlled as

Variable to Be Controlled

Airplane bank angle (denoted by ϕ).

Defining system specifications for aircraft control is complicated, so we do not attempt it here. It is a subject in and of itself, and many engineers have spent significant efforts developing good, practical design specifications. The goal is to design a control system such that the dominant closed-loop system poles have satisfactory natural frequency and damping [24]. We must define satisfactory and choose test input signals on which to base our analysis.

The Cooper–Harper pilot opinion ratings provide a way to correlate the feel of the airplane with control design specifications [26]. These ratings address the handling qualities issues. Many flying qualities requirements are specified by government agencies, such as the United States Air Force [27]. The USAF MIL-F-8785C is a source of time-domain control system design specifications.

For example we might design an autopilot control system for an aircraft in steady, wings-level flight to achieve a $P.O. \leq 20\%$ to a step input with minimal oscillatory motion and rapid response time (that is, a short time-to-peak). Subsequently we implement the controller in the aircraft control system and conduct flight tests or high-fidelity computer simulations, after which the pilots tell us whether they liked the performance of the aircraft. If the overall performance was not satisfactory, we change the time-domain specification (in this case a percent overshoot specification) and redesign until we achieve a feel and performance that pilots (and ultimately passengers) will accept. Despite the simplicity of this approach and many years of research, precise-control system design specifications that provide acceptable airplane flying characteristics in all cases are still not available [24].

The control design specifications given in this example may seem somewhat contrived. In reality the specifications would be much more involved and, in many ways, less precisely known. But we must begin the design process somewhere. With that approach in mind, we select simple design specifications and begin the iterative design process. The design specifications are

Control Design Specifications

DS1 Percent overshoot is $P.O. \leq 20\%$ for a unit step input.

DS2 Fast response time as measured by time-to-peak.

By making the simplifying assumptions discussed above and linearizing about the steady, wings-level flight condition, we can obtain a transfer function model

describing the bank angle output, $\phi(s)$, to the aileron deflection input, $\delta_a(s)$. The transfer function has the form

$$\frac{\phi(s)}{\delta_a(s)} = \frac{k(s - c_0)(s^2 + b_1s + b_0)}{s(s + d_0)(s + e_0)(s^2 + f_1s + f_0)}. \quad (5.69)$$

The lateral (roll/yaw) motion has three main modes: Dutch roll mode, spiral mode, and roll subsidence mode. The Dutch roll mode, which gets its name from its similarities to the motion of an ice speed skater, is characterized by a rolling and yawing motion. The airplane center of mass follows nearly a straightline path, and a rudder impulse can excite this mode. The spiral mode is characterized by a mainly yawing motion with some roll motion. This is a weak mode, but it can cause an airplane to enter a steep spiral dive. The roll subsidence motion is almost a pure roll motion. This is the motion we are concerned with for our autopilot design. The denominator of the transfer function in Equation (5.69) shows two first-order modes (spiral and roll subsidence modes) and a second-order mode (Dutch roll mode).

In general the coefficients $c_0, b_0, b_1, d_0, e_0, f_0, f_1$ and the gain k are complicated functions of stability derivatives. The stability derivatives are functions of the flight conditions and the aircraft configuration; they differ for different aircraft types. The coupling between the roll and yaw is included in Equation (5.69).

In the transfer function in Equation (5.69), the pole at $s = -d_0$ is associated with the spiral mode. The pole at $s = -e_0$ is associated with the roll subsidence mode. Generally, $e_0 \gg d_0$. For an F-16 flying at 500 ft/s in steady, wings-level flight, we have $e_0 = 3.57$ and $d_0 = 0.0128$ [24]. The complex conjugate poles given by the term $s^2 + f_1s + f_0$ represent the Dutch roll motion.

For low angles of attack (such as with steady, wings-level flight), the Dutch roll mode generally cancels out of the transfer function with the $s^2 + b_1s + b_0$ term. This is an approximation, but it is consistent with our other simplifying assumptions. Also, we can ignore the spiral mode since it is essentially a yaw motion only weakly coupled to the roll motion. The zero at $s = c_0$ represents a gravity effect that causes the aircraft to sideslip as it rolls. We assume that this effect is negligible, since it is most pronounced in a slow roll maneuver in which the sideslip is allowed to build up, and we assume that the aircraft sideslip is small or zero. Therefore we can simplify the transfer function in Equation (5.69) to obtain a single-degree-of-freedom approximation:

$$\frac{\phi(s)}{\delta_a(s)} = \frac{k}{s(s + e_0)}. \quad (5.70)$$

For our aircraft we select $e_0 = 1.4$ and $k = 11.4$. The associated time-constant of the roll subsidence is $\tau = 1/e_0 = 0.7$ s. These values represent a fairly fast rolling motion response.

For the aileron actuator model, we typically use a simple first-order system model,

$$\frac{\delta_a(s)}{e(s)} = \frac{p}{s + p}, \quad (5.71)$$

where $e(s) = \phi_d(s) - \phi(s)$. In this case we select $p = 10$. This corresponds to a time constant of $\tau = 1/p = 0.1$ s. This is a typical value consistent with a fast response. We need to have an actuator with a fast response so that the dynamics of the actively controlled airplane will be the dominant component of the system response. A slow actuator is akin to a time delay that can cause performance and stability problems.

For a high-fidelity simulation, we would need to develop an accurate model of the gyro dynamics. The gyro, typically an integrating gyro, is usually characterized by a very fast response. To remain consistent with our other simplifying assumptions, we ignore the gyro dynamics in the design process. This means we assume that the sensor measures the bank angle precisely. The gyro model is given by a unity transfer function,

$$K_g = 1. \quad (5.72)$$

Thus our physical system model is given by Equations (5.70), (5.71), and (5.72).

The controller we select for this design is a proportional controller,

$$G_c(s) = K.$$

The system configuration is shown in Figure 5.30. The select key parameter is as follows:

Select Key Tuning Parameter

Controller gain K .

The closed-loop transfer function is

$$T(s) = \frac{\phi(s)}{\phi_d(s)} = \frac{114K}{s^3 + 11.4s^2 + 14s + 114K}. \quad (5.73)$$

We want to determine analytically the values of K that will give us the desired response, namely, a percent overshoot less than 20% and a fast time-to-peak. The analytic analysis would be simpler if our closed-loop system were a second-order system (since we have valuable relationships between settling time, percent overshoot, natural frequency and damping ratio); however we have a third-order system, given by $T(s)$ in Equation (5.73). We could consider approximating the third-order transfer function by a second-order transfer function—this is sometimes a very good engineering approach to analysis. There are many methods available to

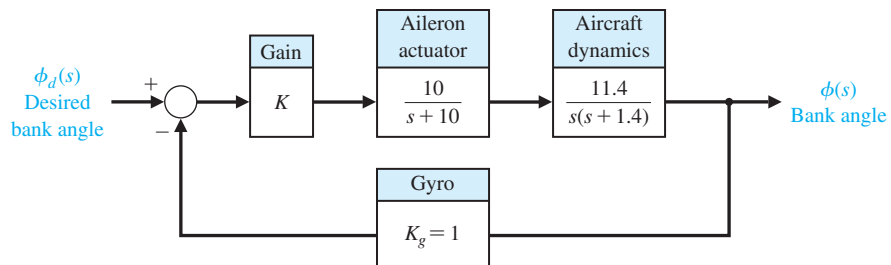


FIGURE 5.30
Bank angle control
autopilot.

obtain approximate transfer functions. Here we use the algebraic method described in Section 5.8 that attempts to match the frequency response of the approximate system as closely as possible to the actual system.

Our transfer function can be rewritten as

$$T(s) = \frac{1}{1 + \frac{14}{114K}s + \frac{11.4}{114K}s^2 + \frac{1}{114K}s^3},$$

by factoring the constant term out of the numerator and denominator. Suppose our approximate transfer function is given by the second-order system

$$G_L(s) = \frac{1}{1 + d_1s + d_2s^2}.$$

The objective is to find appropriate values of d_1 and d_2 . As in Section 5.8, we define $M(s)$ and $\Delta(s)$ as the numerator and denominator of $T(s)/G_L(s)$. We also define

$$M_{2q} = \sum_{k=0}^{2q} \frac{(-1)^{k+q} M^{(k)}(0) M^{(2q-k)}(0)}{k! (2q - k)!}, \quad q = 1, 2, \dots, \quad (5.74)$$

and

$$\Delta_{2q} = \sum_{k=0}^{2q} \frac{(-1)^{k+q} \Delta^{(k)}(0) \Delta^{(2q-k)}(0)}{k! (2q - k)!}, \quad q = 1, 2, \dots \quad (5.75)$$

Then, forming the set of algebraic equations

$$M_{2q} = \Delta_{2q}, \quad q = 1, 2, \dots, \quad (5.76)$$

we can solve for the unknown parameters of the approximate function. The index q is incremented until sufficient equations are obtained to solve for the unknown coefficients of the approximate function. In this case, $q = 1, 2$ since we have two parameters d_1 and d_2 to compute.

We have

$$\begin{aligned} M(s) &= 1 + d_1s + d_2s^2 \\ M^{(1)}(s) &= \frac{dM}{ds} = d_1 + 2d_2s \\ M^{(2)}(s) &= \frac{d^2M}{ds^2} = 2d_2 \\ M^{(3)}(s) &= M^{(4)}(s) = \dots = 0. \end{aligned}$$

Thus evaluating at $s = 0$ yields

$$\begin{aligned} M^{(1)}(0) &= d_1 \\ M^{(2)}(0) &= 2d_2 \\ M^{(3)}(0) &= M^{(4)}(0) = \dots = 0. \end{aligned}$$

Similarly,

$$\begin{aligned}\Delta(s) &= 1 + \frac{14}{114K}s + \frac{11.4}{114K}s^2 + \frac{s^3}{114K} \\ \Delta^{(1)}(s) &= \frac{d\Delta}{ds} = \frac{14}{114K} + \frac{22.8}{114K}s + \frac{3}{114K}s^2 \\ \Delta^{(2)}(s) &= \frac{d^2\Delta}{ds^2} = \frac{22.8}{114K} + \frac{6}{114K}s \\ \Delta^{(3)}(s) &= \frac{d^3\Delta}{ds^3} = \frac{6}{114K} \\ \Delta^{(4)}(s) &= \Delta^{(5)}(s) = \dots = 0.\end{aligned}$$

Evaluating at $s = 0$, it follows that

$$\begin{aligned}\Delta^{(1)}(0) &= \frac{14}{114K}, \\ \Delta^{(2)}(0) &= \frac{22.8}{114K}, \\ \Delta^{(3)}(0) &= \frac{6}{114K}, \\ \Delta^{(4)}(0) &= \Delta^{(5)}(0) = \dots = 0.\end{aligned}$$

Using Equation (5.77) for $q = 1$ and $q = 2$ yields

$$M_2 = -\frac{M(0)M^{(2)}(0)}{2} + \frac{M^{(1)}(0)M^{(1)}(0)}{1} - \frac{M^{(2)}(0)M(0)}{2} = -2d_2 + d_1^2,$$

and

$$\begin{aligned}M_4 &= \frac{M(0)M^{(4)}(0)}{0!4!} - \frac{M^{(1)}(0)M^{(3)}(0)}{1!3!} + \frac{M^{(2)}(0)M^{(2)}(0)}{2!2!} \\ &\quad - \frac{M^{(3)}(0)M^{(1)}(0)}{3!1!} + \frac{M^{(4)}(0)M(0)}{4!0!} = d_2^2.\end{aligned}$$

Similarly using Equation (5.78), we find that

$$\Delta_2 = \frac{-22.8}{114K} + \frac{196}{(114K)^2} \quad \text{and} \quad \Delta_4 = \frac{101.96}{(114K)^2}.$$

Thus forming the set of algebraic equations in Equation (5.79),

$$M_2 = \Delta_2 \quad \text{and} \quad M_4 = \Delta_4,$$

we obtain

$$-2d_2 + d_1^2 = \frac{-22.8}{114K} + \frac{196}{(114K)^2} \quad \text{and} \quad d_2^2 = \frac{101.96}{(114K)^2}.$$

Solving for d_1 and d_2 yields

$$d_1 = \frac{\sqrt{196 - 296.96K}}{114K}, \quad (5.77)$$

$$d_2 = \frac{10.097}{114K}, \quad (5.78)$$

where we always choose the positive values of d_1 and d_2 so that $G_L(s)$ has poles in the left half-plane. Thus (after some manipulation) the approximate transfer function is

$$G_L(s) = \frac{11.29K}{s^2 + \sqrt{1.92 - 2.91K}s + 11.29K}. \quad (5.79)$$

We require that $K < 0.65$ so that the coefficient of the s term remains a real number. Our desired second-order transfer function can be written as

$$G_L(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5.80)$$

Comparing coefficients in Equations (5.79) and (5.80) yields

$$\omega_n^2 = 11.29K \quad \text{and} \quad \zeta^2 = \frac{0.043}{K} - 0.065. \quad (5.81)$$

The design specification that the percent overshoot $P.O.$ is to be less than 20% implies that we want $\zeta \geq 0.45$. Setting $\zeta = 0.45$ in Equation (5.81) and solving for K yields

$$K = 0.16.$$

With $K = 0.16$ we compute

$$\omega_n = \sqrt{11.29K} = 1.34.$$

Then we can estimate the time-to-peak T_p from Equation (5.14) to be

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = 2.62\text{s}.$$

We might be tempted at this point to select $\zeta > 0.45$ so that we reduce the percent overshoot even further than 20%. What happens if we decide to try this approach? From Equation (5.81) we see that K decreases as ζ increases. Then, since

$$\omega_n = \sqrt{11.29K},$$

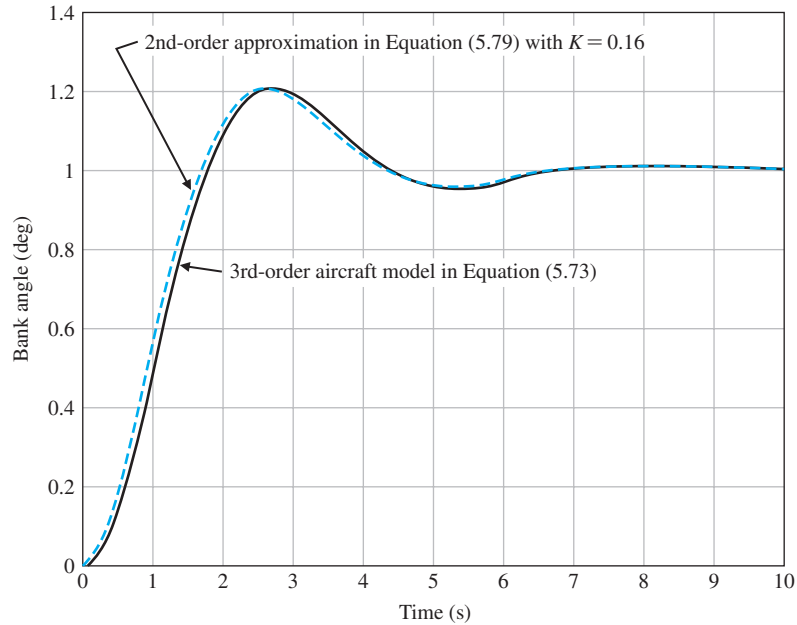


FIGURE 5.31 Step response comparison of third-order aircraft model versus second-order approximation.

as K decreases, then ω_n also decreases. But the time-to-peak increases as ω_n decreases. Since our goal is to meet the specification of percent overshoot less than 20% while minimizing the time-to-peak, we use the initial selection of $\zeta = 0.45$ so that we do not increase T_p unnecessarily.

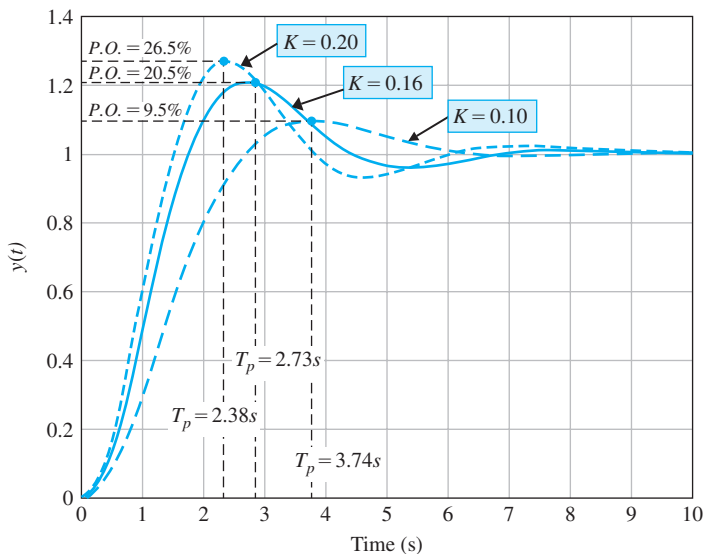
The second-order system approximation has allowed us to gain insight into the relationship between the parameter K and the system response, as measured by percent overshoot and time-to-peak. Of course, the gain $K = 0.16$ is only a starting point in the design because we in fact have a third-order system and must consider the effect of the third pole (which we have ignored so far).

A comparison of the third-order aircraft model in Equation (5.73) with the second-order approximation in Equation (5.79) for a unit step input is shown in Figure 5.31. The step response of the second-order system is a good approximation of the original system step response, so we would expect that the analytic analysis using the simpler second-order system to provide accurate indications of the relationship between K and the percent overshoot and time-to-peak.

With the second-order approximation, we estimate that with $K = 0.16$ the percent overshoot is $P.O. = 20\%$ and the time-to-peak is $T_p = 2.62$ s. As shown in Figure 5.32 the percent overshoot of the original third-order system is $P.O. = 20.5\%$ and the time-to-peak is $T_p = 2.73$ s. Thus, we see that that analytic analysis using the approximate system is an excellent predictor of the actual response. For comparison purposes, we select two variations in the gain and observe the response. For $K = 0.1$, the percent overshoot is $P.O. = 9.5\%$ and the time-to-peak is $T_p = 3.74$ s.

FIGURE 5.32

Step response of the third-order aircraft model with $K = 0.10, 0.16$, and 0.20 showing that, as predicted, as K decreases percent overshoot decreases while the time-to-peak increases.



For $K = 0.2$, the percent overshoot is $P.O. = 26.5\%$ and the time-to-peak is $T_p = 2.38s$. So as predicted, as K decreases the damping ratio increases, leading to a reduction in the percent overshoot. Also as predicted, as the percent overshoot decreases the time-to-peak increases. ■

5.10 SYSTEM PERFORMANCE USING CONTROL DESIGN SOFTWARE

In this section, we investigate time-domain performance specifications given in terms of transient response to a given input signal and the resulting steady-state tracking errors. We conclude with a discussion of the simplification of linear systems. The function introduced in this section is **impulse**. We discuss the **lsim** function and see how these functions are used to simulate a linear system.

Time-Domain Specifications. Time-domain performance specifications are generally given in terms of the transient response of a system to a given input signal. Because the actual input signals are generally unknown, a standard test input signal is used. Consider the second-order system shown in Figure 5.3. The closed-loop output is

$$Y(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s). \quad (5.82)$$

We have already discussed the use of the **step** function to compute the step response of a system. Now we address another important test signal: the impulse.

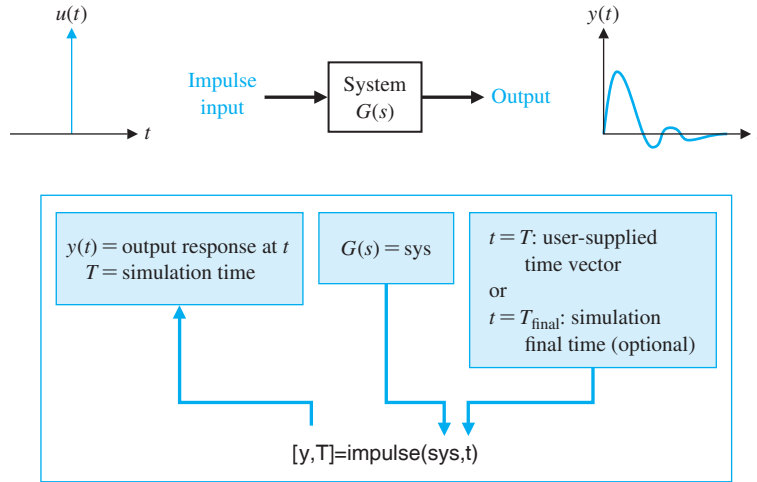


FIGURE 5.33 The **impulse** function.

The impulse response is the time derivative of the step response. We compute the impulse response with the **impulse** function shown in Figure 5.33.

We can obtain a plot similar to that of Figure 5.4 with the **step** function, as shown in Figure 5.34. Using the **impulse** function, we can obtain a plot similar to that of Figure 5.5. The response of a second-order system for an impulse function input is shown in Figure 5.35. In the script, we set $\omega_n = 1$, which is equivalent to computing the step response versus $\omega_n t$. This gives us a more general plot valid for any $\omega_n > 0$.

In many cases, it may be necessary to simulate the system response to an arbitrary but known input. In these cases, we use the **lsim** function. The **lsim** function is shown in Figure 5.36.

EXAMPLE 5.10 Mobile robot steering control

The block diagram for a steering control system for a mobile robot is shown in Figure 5.18. Suppose the transfer function of the steering controller is

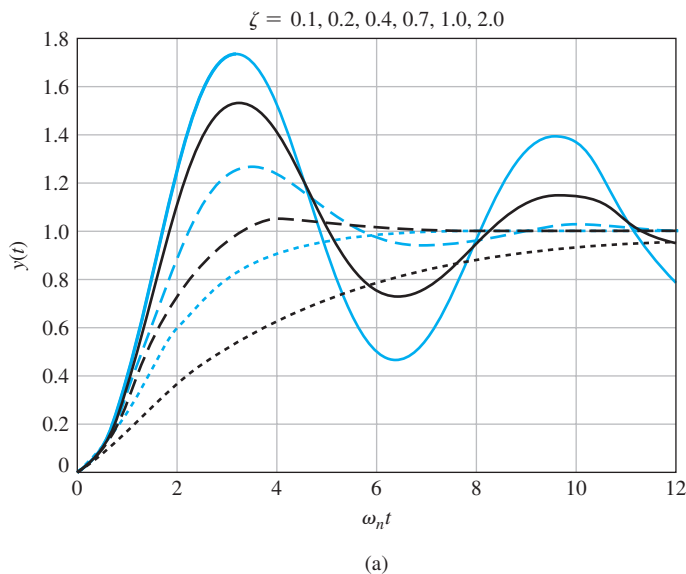
$$G_c(s) = K_1 + \frac{K_2}{s}.$$

When the input is a ramp, the steady-state error is

$$e_{ss} = \frac{A}{K_v}, \quad (5.83)$$

where

$$K_v = K_2 K.$$



```
%Compute step response for a second-order system
%Duplicate Figure 5.4
%
t=[0:0.1:12]; num=[1];
zeta1=0.1; den1=[1 2*zeta1 1]; sys1=tf(num,den1);
zeta2=0.2; den2=[1 2*zeta2 1]; sys2=tf(num,den2);
zeta3=0.4; den3=[1 2*zeta3 1]; sys3=tf(num,den3);
zeta4=0.7; den4=[1 2*zeta4 1]; sys4=tf(num,den4);
zeta5=1.0; den5=[1 2*zeta5 1]; sys5=tf(num,den5);
zeta6=2.0; den6=[1 2*zeta6 1]; sys6=tf(num,den6);
%
[y1,T1]=step(sys1,t); [y2,T2]=step(sys2,t);
[y3,T3]=step(sys3,t); [y4,T4]=step(sys4,t);
[y5,T5]=step(sys5,t); [y6,T6]=step(sys6,t);
%
plot(T1,y1,T2,y2,T3,y3,T4,y4,T5,y5,T6,y6)
xlabel('ω_n t'), ylabel('y(t)')
title('ζ = 0.1, 0.2, 0.4, 0.7, 1.0, 2.0'), grid
```

Compute
step
response.

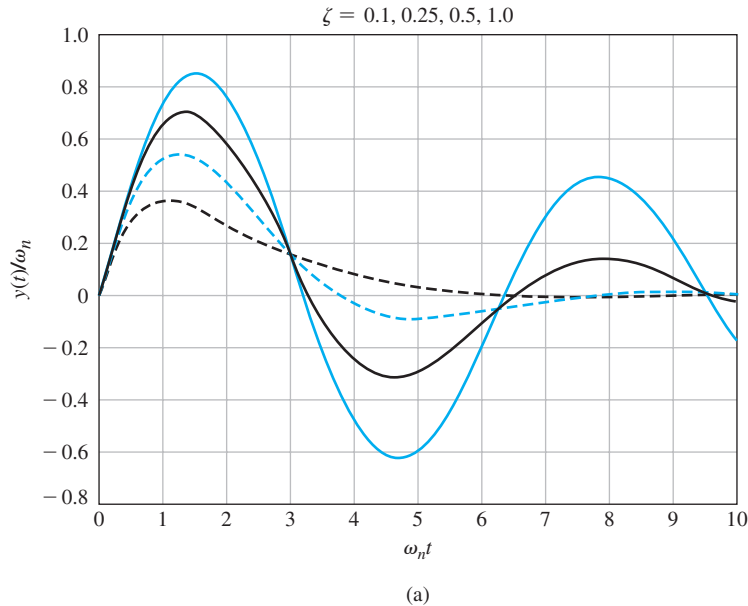
Generate plot
and labels.

FIGURE 5.34
(a) Response of
a second-order
system to a step
input. (b) m-file
script.

(b)

The effect of the controller constant, K_2 , on the steady-state error is evident from Equation (5.83). Whenever K_2 is large, the steady-state error is small.

We can simulate the closed-loop system response to a ramp input using the `lsim` function. The controller gains, K_1 and K_2 , and the system gain K can be represented symbolically in the script so that various values can be selected and simulated. The results are shown in Figure 5.37 for $K_1 = K = 1$, $K_2 = 2$, and $\tau = 1/10$. ■



```
%Compute impulse response for a second-order system
%Duplicate Figure 5.5
%
t=[0:0.1:10]; num=[1];
zeta1=0.1; den1=[1 2*zeta1 1]; sys1=tf(num,den1);
zeta2=0.25; den2=[1 2*zeta2 1]; sys2=tf(num,den2);
zeta3=0.5; den3=[1 2*zeta3 1]; sys3=tf(num,den3);
zeta4=1.0; den4=[1 2*zeta4 1]; sys4=tf(num,den4);
%
[y1,T1]=impz(sys1,t);
[y2,T2]=impz(sys2,t);
[y3,T3]=impz(sys3,t);
[y4,T4]=impz(sys4,t);
%
plot(t,y1,t,y2,t,y3,t,y4)
xlabel('omega_n t'), ylabel('y(t)/omega_n')
title('\zeta = 0.1, 0.25, 0.5, 1.0'), grid
```

← Compute impulse response.

← Generate plot and labels.

FIGURE 5.35
(a) Response of a second-order system to an impulse function input. (b) m-file script.

(b)

Simplification of Linear Systems. It may be possible to develop a lower-order approximate model that closely matches the input–output response of a high-order model. A procedure for approximating transfer functions is given in Section 5.8. We can use computer simulation to compare the approximate model to the actual model, as illustrated in the following example.

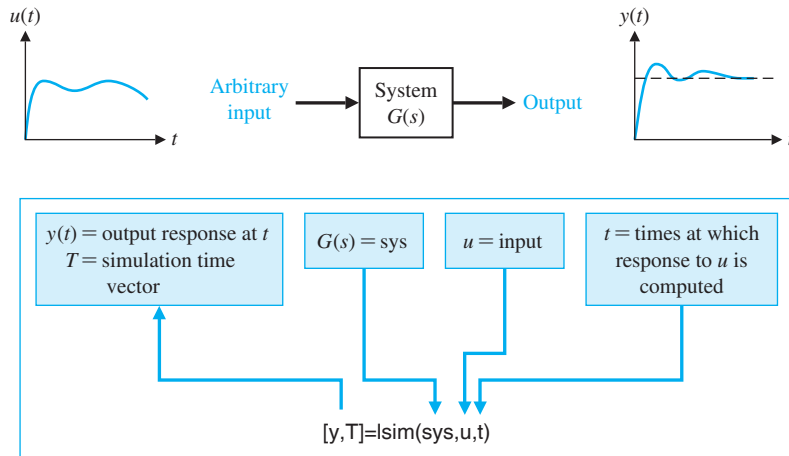


FIGURE 5.36
The `lsim` function.

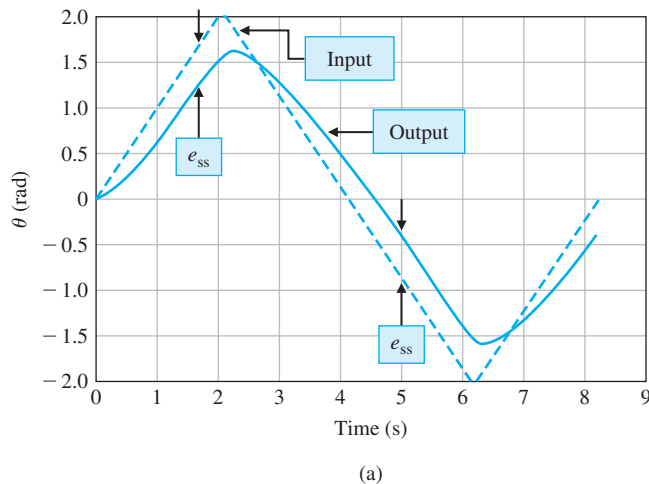


FIGURE 5.37
(a) Transient response of the mobile robot steering control system to a ramp input. (b) m-file script.

```
%Compute the response of the Mobile Robot Control
%System to a triangular wave input
%
numg=[10 20]; deng=[1 10 0]; sysg=tf(numg,deng); % G(s)G_c(s)
[sys]=feedback(sysg, [1]);
t=[0:0.1:8.2]';
v1=[0:0.1:2]'; v2=[2:-0.1:-2]'; v3=[-2:0.1:0]'; % Compute triangular wave input.
u=[v1;v2;v3];
[y,T]=lsim(sys,u,t); % Linear simulation.
plot(T,y,t,u,'--'),
xlabel('Time (s)'), ylabel('\theta (rad)'), grid
```

(b)

EXAMPLE 5.11 A simplified model

Consider the third-order system

$$G_H(s) = \frac{6}{s^3 + 6s^2 + 11s + 6}.$$

A second-order approximation (see Example 5.7) is

$$G_L(s) = \frac{1.60}{s^2 + 2.590s + 1.60}.$$

A comparison of their respective step responses is given in Figure 5.38. ■

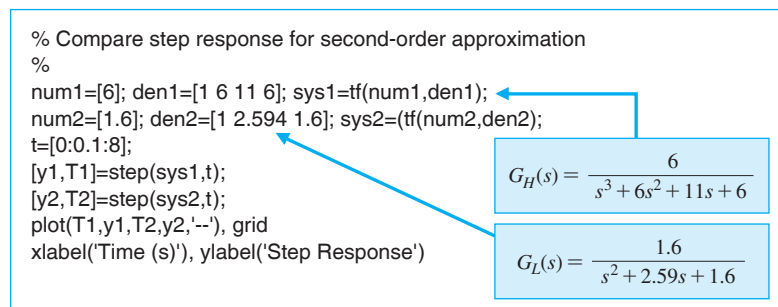
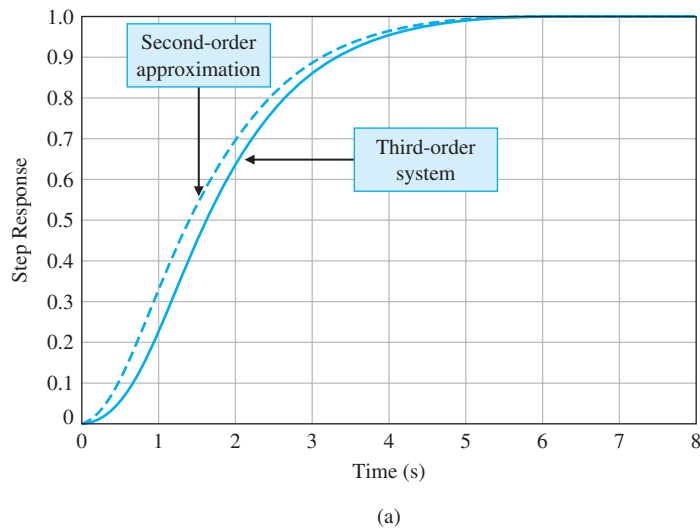


FIGURE 5.38 (a) Step response comparison for an approximate transfer function versus the actual transfer function. (b) m-file script.

5.11 SEQUENTIAL DESIGN EXAMPLE: DISK DRIVE READ SYSTEM

In this section, we further consider the design process of the disk drive read system. We will specify the desired performance for the system. Then we will attempt to adjust the amplifier gain K_a in order to obtain the best performance possible.

Our goal is to achieve the fastest response to a step input $r(t)$ while (1) limiting the percent overshoot and oscillatory nature of the response and (2) reducing the effect of a disturbance on the output position of the read head. The specifications are summarized in Table 5.5.

Let us consider the second-order model of the motor and arm, which neglects the effect of the coil inductance. We then have the closed-loop system shown in Figure 5.39. Then the output when $T_d(s) = 0$ is

$$\begin{aligned} Y(s) &= \frac{5K_a}{s(s+20) + 5K_a} R(s) \\ &= \frac{5K_a}{s^2 + 20s + 5K_a} R(s) \\ &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s). \end{aligned} \quad (5.84)$$

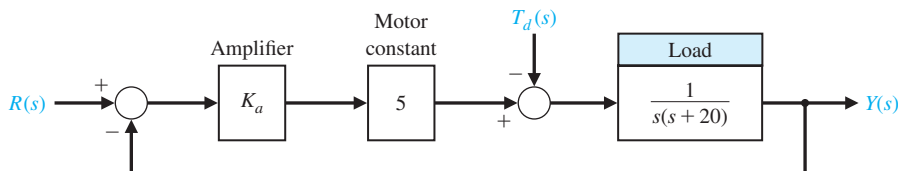
Therefore, $\omega_n^2 = 5K_a$, and $2\zeta\omega_n = 20$. We then determine the response of the system as shown in Figure 5.40. Table 5.6 shows the performance measures for selected values of K_a .

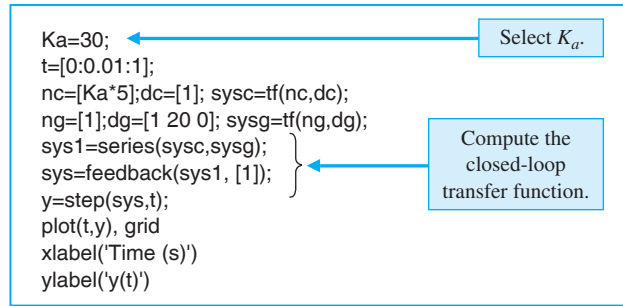
When K_a is increased to 60, the effect of a disturbance is reduced by a factor of 2. We can show this by plotting the output, $y(t)$, as a result of a unit step disturbance

Table 5.5 Specifications for the Transient Response

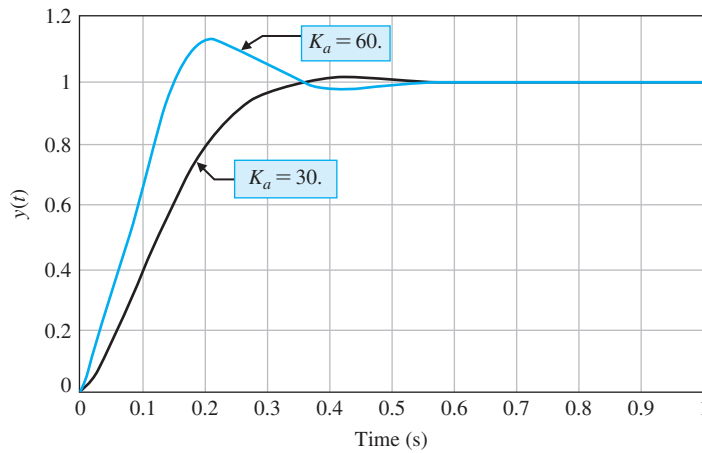
Performance Measure	Desired Value
Percent overshoot	Less than 5%
Settling time	Less than 250 ms
Maximum value of response to a unit step disturbance	Less than 5×10^{-3}

FIGURE 5.39
Control system model with a second-order model of the motor and load.





(a)



(b)

FIGURE 5.40
 Response of the system to a unit step input, $r(t) = 1, t > 0$.
 (a) m-file script.
 (b) Response for $K_a = 30$ and 60.

Table 5.6 Response for the Second-Order Model for a Step Input

K_a	20	30	40	60	80
Percent overshoot	0	1.2%	4.3%	10.8%	16.3%
Settling time (s)	0.55	0.40	0.40	0.40	0.40
Damping ratio	1	0.82	0.707	0.58	0.50
Maximum value of the response $y(t)$ to a unit disturbance	-10×10^{-3}	-6.6×10^{-3}	-5.2×10^{-3}	-3.7×10^{-3}	-2.9×10^{-3}

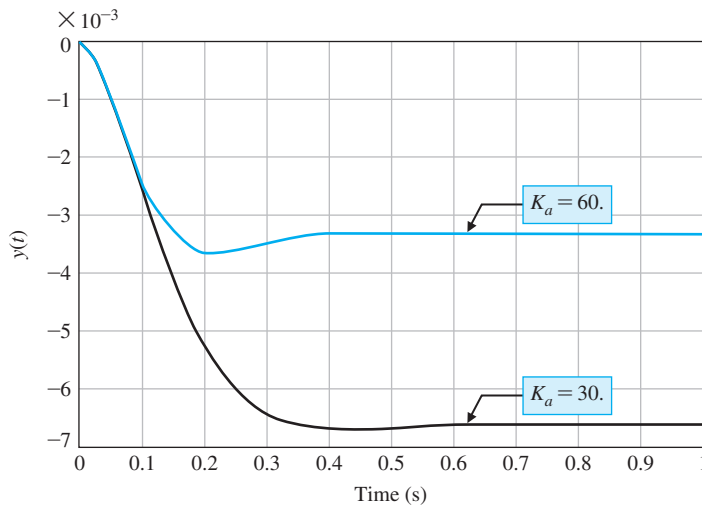
input, as shown in Figure 5.41. Clearly, if we wish to meet our goals with this system, we need to select a compromise gain. In this case, we select $K_a = 40$ as the best compromise. However, this compromise does not meet all the specifications. In the next chapter, we consider again the design process and change the configuration of the control system.

```

Ka=30; ← Select  $K_a$ .
t=[0:0.01:1];
nc=[Ka*5];dc=[1]; sysc=tf(nc,dc);
ng=[1];dg=[1 20 0]; sysg=tf(ng,dg);
sys=feedback(sysg,sysc);
sys=-sys; ← Disturbance enters summer
           with a negative sign.
y=step(sys,t); plot(t,y)
xlabel('Time (s)'), ylabel('y(t)'), grid

```

(a)



(b)

FIGURE 5.41
 Response of the system to a unit step disturbance, $T_d(s) = 1/s$.
 (a) m-file script.
 (b) Response for $K_a = 30$ and 60 .

5.12 SUMMARY

In this chapter, we have considered the definition and measurement of the performance of a feedback control system. The concept of a performance measure or index was discussed, and the usefulness of standard test signals was outlined. Then, several performance measures for a standard step input test signal were delineated. For example, the overshoot, peak time, and settling time of the response of the system under test for a step input signal were considered. The fact that the specifications on the desired response are often contradictory was noted, and the concept of a design compromise was proposed. The relationship between the location of the s -plane root of the system transfer function and the system response was discussed. A most important measure of system performance is the steady-state error for specific test input signals. Thus, the relationship of the steady-state error of a system in terms of the system parameters was developed by utilizing the final-value theorem. Finally, the utility of an integral performance index was outlined, and several design examples that minimized a system's performance index were completed. Thus, we have been concerned with the definition and usefulness of quantitative measures of the performance of feedback control systems.

SKILLS CHECK

In this section, we provide three sets of problems to test your knowledge: True or False, Multiple Choice, and Word Match. To obtain direct feedback, check your answers with the answer key provided at the conclusion of the end-of-chapter problems. Use the block diagram in Figure 5.42 as specified in the various problem statements.

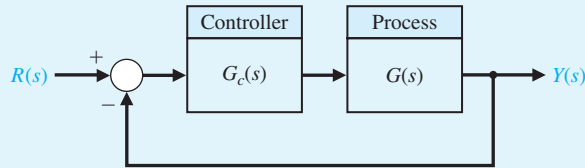


FIGURE 5.42 Block diagram for the Skills Check.

In the following **True or False** and **Multiple Choice** problems, circle the correct answer.

1. In general, a third-order system can be approximated by a second-order system's dominant roots if the real part of the dominant roots is less than 1/10 of the real part of the third root. *True or False*
2. The number of zeros of the forward path transfer function at the origin is called the type number. *True or False*
3. The rise time is defined as the time required for the system to settle within a certain percentage of the input amplitude. *True or False*
4. For a second-order system with no zeros, the percent overshoot to a unit step is a function of the damping ratio only. *True or False*
5. A type-1 system has a zero steady-state tracking error to a ramp input. *True or False*

Consider the closed-loop control system in Figure 5.42 for Problems 6 and 7 with

$$L(s) = G_c(s)G(s) = \frac{6}{s(s+3)}.$$

6. The steady-state error to a unit step input $R(s) = 1/s$ is:
 - a. $e_{ss} = \lim_{t \rightarrow \infty} e(t) = 1$
 - b. $e_{ss} = \lim_{t \rightarrow \infty} e(t) = 1/2$
 - c. $e_{ss} = \lim_{t \rightarrow \infty} e(t) = 1/6$
 - d. $e_{ss} = \lim_{t \rightarrow \infty} e(t) = \infty$
7. The percent overshoot of the output to a unit step input is:
 - a. $P.O. = 9\%$
 - b. $P.O. = 1\%$
 - c. $P.O. = 20\%$
 - d. No overshoot

Consider the block diagram of the control system shown in Figure 5.42 in Problems 8 and 9 with the loop transfer function

$$L(s) = G_c(s)G(s) = \frac{K}{s(s+10)}.$$

8. Find the value of K so that the system provides an optimum ITAE response.
 - a. $K = 1.10$
 - b. $K = 12.56$
 - c. $K = 51.02$
 - d. $K = 104.7$
9. Compute the expected percent overshoot to a unit step input.
 - a. $P.O. = 1.4\%$
 - b. $P.O. = 4.6\%$
 - c. $P.O. = 10.8\%$
 - d. No overshoot expected

10. A system has the closed-loop transfer function $T(s)$ given by

$$T(s) = \frac{Y(s)}{R(s)} = \frac{2500}{(s + 20)(s^2 + 10s + 125)}.$$

Using the notion of dominant poles, estimate the expected percent overshoot.

- a. $P.O. \approx 5\%$
 - b. $P.O. \approx 20\%$
 - c. $P.O. \approx 50\%$
 - d. No overshoot expected
11. Consider the unity feedback control system in Figure 5.42 where

$$L(s) = G_c(s)G(s) = \frac{K}{s(s + 5)}.$$

The design specifications are:

- i. Peak time $T_p \leq 1.0$
- ii. Percent overshoot $P.O. \leq 10\%$.

With K as the design parameter, it follows that

- a. Both specifications can be satisfied.
 - b. Only the first specification $T_p \leq 1.0$ can be satisfied.
 - c. Only the second specification $P.O. \leq 10\%$ can be satisfied.
 - d. Neither specification can be satisfied.
12. Consider the feedback control system in Figure 5.43 where $G(s) = \frac{K}{s + 10}$.

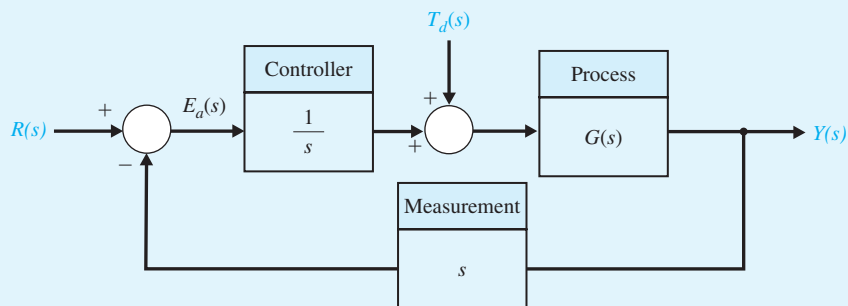


FIGURE 5.43 Feedback system with integral controller and derivative measurement.

The nominal value of $K = 10$. Using a 2% criterion, compute the settling time, T_s for a unit step disturbance, $T_d(s) = 1/s$.

- a. $T_s = 0.02$ s
- b. $T_s = 0.19$ s
- c. $T_s = 1.03$ s
- d. $T_s = 4.83$ s

13. A plant has the transfer function given by

$$G(s) = \frac{1}{(1 + s)(1 + 0.5s)}$$

and is controlled by a proportional controller $G_c(s) = K$, as shown in the block diagram in Figure 5.42. The value of K that yields a steady-state error $E(s) = Y(s) - R(s)$ with a magnitude equal to 0.01 for a unit step input is:

- a. $K = 49$
- b. $K = 99$
- c. $K = 169$
- d. None of the above

In Problems 14 and 15, consider the control system in Figure 5.42, where

$$G(s) = \frac{6}{(s + 5)(s + 2)} \quad \text{and} \quad G_c(s) = \frac{K}{s + 50}.$$

14. A second-order approximate model of the loop transfer function is:

- a. $\hat{G}_c(s)\hat{G}(s) = \frac{(3/25)K}{s^2 + 7s + 10}$
- b. $\hat{G}_c(s)\hat{G}(s) = \frac{(1/25)K}{s^2 + 7s + 10}$
- c. $\hat{G}_c(s)\hat{G}(s) = \frac{(3/25)K}{s^2 + 7s + 500}$
- d. $\hat{G}_c(s)\hat{G}(s) = \frac{6K}{s^2 + 7s + 10}$

15. Using the second-order system approximation (see Problem 14), estimate the gain K so that the percent overshoot is approximately $P.O. \approx 15\%$.

- a. $K = 10$
- b. $K = 300$
- c. $K = 1000$
- d. None of the above

In the following **Word Match** problems, match the term with the definition by writing the correct letter in the space provided.

- | | | |
|------------------|--|-------|
| a. Unit impulse | The time for a system to respond to a step input and rise to a peak response. | _____ |
| b. Rise time | The roots of the characteristic equation that cause the dominant transient response of the system. | _____ |
| c. Settling time | The number N of poles of the transfer function, $G(s)$, at the origin. | _____ |

d. Type number	The constant evaluated as $\lim_{s \rightarrow 0} s G(s)$.	_____
e. Percent overshoot	An input signal used as a standard test of a system's ability to respond adequately.	_____
f. Position error constant, K_p	The time required for the system output to settle within a certain percentage of the input amplitude.	_____
g. Velocity error constant, K_v	A set of prescribed performance criteria.	_____
h. Steady-state response	A system whose parameters are adjusted so that the performance index reaches an extremum value.	_____
i. Peak time	A quantitative measure of the performance of a system.	_____
j. Dominant roots	The time for a system to respond to a step input and attain a response equal to a percentage of the magnitude of the input.	_____
k. Test input signal	The amount by which the system output response proceeds beyond the desired response.	_____
l. Acceleration error constant, K_a	The constant evaluated as $\lim_{s \rightarrow 0} s^2 G(s)$.	_____
m. Transient response	The constant evaluated as $\lim_{s \rightarrow 0} G(s)$.	_____
n. Design specifications	The constituent of the system response that exists a long time following any signal initiation.	_____
o. Performance index	The constituent of the system response that disappears with time.	_____
p. Optimum control system	A test input consisting of an impulse of infinite amplitude and zero width, and having an area of unity.	_____

EXERCISES

E5.1 A motor control system for a computer disk drive must reduce the effect of disturbances and parameter variations, as well as reduce the steady-state error. We want to have no steady-state error for the head-positioning control system. (a) What type number is required? (How many integrations?) (b) If the input is a ramp signal, and we want to achieve a zero steady-state error, what type number is required?

E5.2 The engine, body, and tires of a racing vehicle affect the acceleration and speed attainable [9]. The speed control of the car is represented by the model shown in Figure E5.2. (a) Calculate the steady-state error of the car to a step command in speed. (b) Calculate percent overshoot of the speed to a step command.
Answer: (a) $e_{ss} = A/5.2$; (b) $P.O. = 21\%$

E5.3 New passenger rail systems that could profitably compete with air travel are under development. Two of these systems, the French TGV and the Japanese Shinkansen, reach speeds of 160 mph [17]. The Trans-rapid, a magnetic levitation train, is shown in Figure E5.3(a).

The use of magnetic levitation and electro-magnetic propulsion to provide contactless vehicle

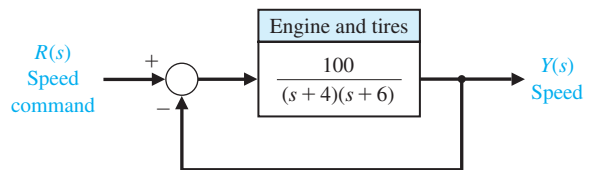


FIGURE E5.2 Racing car speed control.

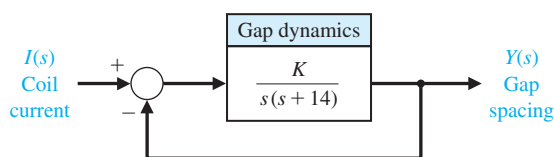
movement makes the Transrapid technology radically different. The underside of the carriage (where the wheel trucks would be on a conventional car) wraps around a guideway. Magnets are attached to the wraparound and pull the train to the reaction rail at the bottom of the guideway.

The levitation control is represented by Figure E5.3(b). (a) Select K so that the system provides an optimum ITAE response. (b) Determine the expected percent overshoot to a step input of $I(s)$.

Answer: $K = 100$; 4.6%



(a)



(b)

FIGURE E5.3 Levitated train control. (Part a courtesy of ThyssenKrupp Transrapid GmbH.)

E5.4 A feedback system with negative unity feedback has a loop transfer function

$$L(s) = G_c(s)G(s) = \frac{2(s + 8)}{s(s + 4)}.$$

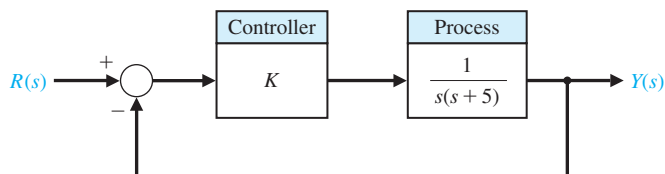
(a) Determine the closed-loop transfer function $T(s) = Y(s)/R(s)$. (b) Find the time response, $y(t)$, for a step input $r(t) = A$ for $t > 0$. (c) Determine the percent overshoot of the response. (d) Using the final-value theorem, determine the steady-state value of $y(t)$.

Answer: (b) $y(t) = 1 - 1.07e^{-3t} \sin(\sqrt{7}t + 1.2)$

E5.5 Consider the feedback system in Figure E5.5. Find K such that the closed-loop system minimizes the ITAE performance criterion for a step input.

E5.6 Consider the block diagram shown in Figure E5.6 [16]. (a) Calculate the steady-state error for a ramp input. (b) Select a value of K that will result in zero percent overshoot to a step input. Provide rapid response.

FIGURE E5.5 Feedback system with proportional controller $G_c(s) = K$.



Plot the poles and zeros of this system and discuss the dominance of the complex poles. What overshoot for a step input do you expect?

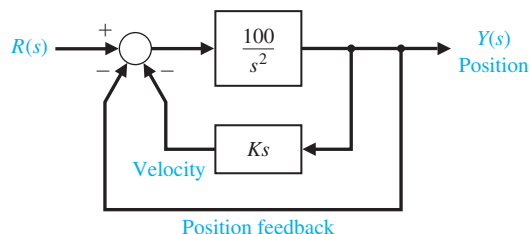


FIGURE E5.6 Block diagram with position and velocity feedback.

E5.7 Effective control of insulin injections can result in better lives for diabetic persons. Automatically controlled insulin injection by means of a pump and a sensor that measures blood sugar can be very effective. A pump and injection system has a feedback control as shown in Figure E5.7. Calculate the suitable gain K so that the percent overshoot of the step response due to the drug injection is $P.O. = 7\%$. $R(s)$ is the desired blood-sugar level and $Y(s)$ is the actual blood-sugar level.

Answer: $K = 1.67$

E5.8 A control system for positioning the head of a floppy disk drive has the closed-loop transfer function

$$T(s) = \frac{11.1(s + 18)}{(s + 20)(s^2 + 4s + 10)}.$$

Plot the poles and zeros of this system and discuss the dominance of the complex poles. What percent overshoot for a step input do you expect?

E5.9 A unity negative feedback control system has the loop transfer function

$$L(s) = G_c(s)G(s) = \frac{K}{s(s + \sqrt{2K})}.$$

(a) Determine the percent overshoot and settling time (using a 2% settling criterion) due to a unit step input.

(b) For what range of K is the settling time is $T_s \leq 1$ s?

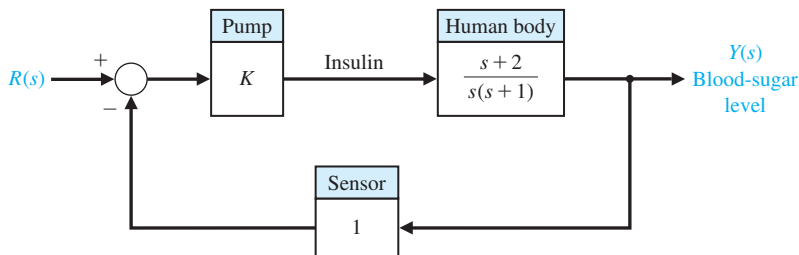


FIGURE E5.7
Blood-sugar level control.

E5.10 A second-order control system has the closed-loop transfer function $T(s) = Y(s)/R(s)$. The system specifications for a step input follow:

1. Percent overshoot $P.O. \leq 5\%$.
2. Settling time $T_s < 4s$.
3. Peak time $T_p < 1s$.

Show the desired region for the poles of $T(s)$ in order to achieve the desired response. Use a 2% settling criterion to determine settling time.

E5.11 A system with unity feedback is shown in Figure E5.11. Determine the steady-state error for a step and a ramp input when

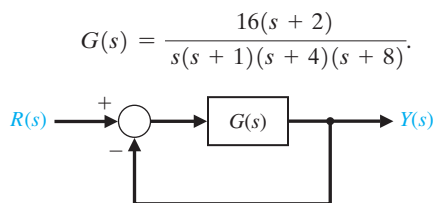


FIGURE E5.11 Unity feedback system.

E5.12 The Ferris wheel is often featured at state fairs and carnivals. George Ferris was born in Galesburg, Illinois, in 1859; he later moved to Nevada and then graduated from Rensselaer Polytechnic Institute in 1881. By 1891, Ferris had considerable experience with iron, steel, and bridge construction. He conceived and constructed his famous wheel for the 1893 Columbian Exposition in Chicago [8]. Consider the requirement that the steady-state speed must be controlled to within 5% of the desired speed for the Ferris wheel speed control system shown in Figure E5.12.

- (a) Determine the required gain K to achieve the steady-state requirement.

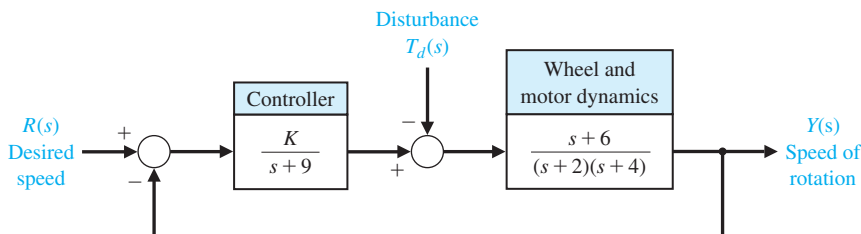


FIGURE E5.12
Speed control of a Ferris wheel.

- (b) For the gain of part (a), determine and plot the tracking error for a unit step disturbance. Does the speed change more than 5%? (Set $R(s) = 0$ and recall that the tracking error $E(s) = R(s) - T(s)$.)

E5.13 For the system with unity feedback shown in Figure E5.11, determine the steady-state error for a step and a ramp input when

$$G(s) = \frac{20}{s^2 + 14s + 50}.$$

Answer: $e_{ss} = 0.71$ for a step and $e_{ss} = \infty$ for a ramp.

E5.14 A feedback system is shown in Figure E5.14.

- (a) Determine the steady-state error for a unit step when $K = 0.4$ and $G_p(s) = 1$.
- (b) Select an appropriate value for $G_p(s)$ so that the steady-state error is equal to zero for the unit step input.

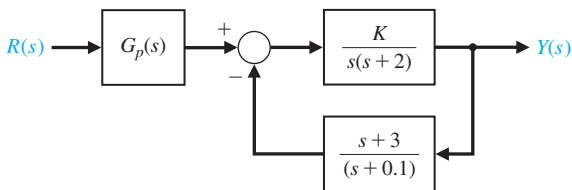


FIGURE E5.14 Feedback system.

E5.15 A closed-loop control system has a transfer function $T(s)$ as follows:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{2500}{(s+50)(s^2+10s+50)}.$$

Plot $y(t)$ for a unit step input when (a) the actual $T(s)$ is used, and (b) using the dominant complex poles. Compare the results.

E5.16 A second-order system is

$$T(s) = \frac{Y(s)}{R(s)} = \frac{(10/z)(s+z)}{(s+1)(s+8)}.$$

Consider the case where $1 < z < 8$. Obtain the partial fraction expansion, and plot the output for a unit step input for $z = 2, 4$, and 6 .

E5.17 A closed-loop control system transfer function $T(s)$ has two dominant complex conjugate poles. Sketch the region in the left-hand s -plane where the complex poles should be located to meet the given specifications.

- (a) $0.6 \leq \zeta \leq 0.8$, $\omega_n \leq 10$
- (b) $0.5 \leq \zeta \leq 0.707$, $\omega_n \geq 10$
- (c) $\zeta \geq 0.5$, $5 \leq \omega_n \leq 10$
- (d) $\zeta \leq 0.707$, $5 \leq \omega_n \leq 10$
- (e) $\zeta \geq 0.6$, $\omega_n \leq 6$

E5.18 A system is shown in Figure E5.18(a). The response to a unit step, when $K = 1$, is shown in Figure E5.18(b). Determine the value of K so that the steady-state error is equal to zero.

Answer: $K = 1.25$.

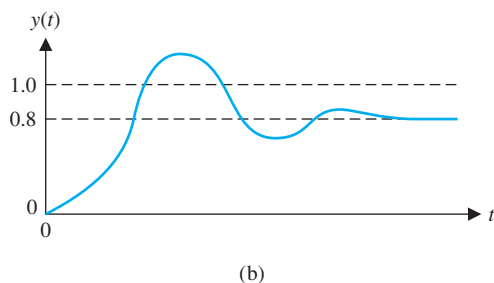
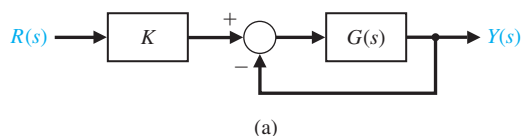


FIGURE E5.18 Feedback system with prefilter.

PROBLEMS

P5.1 An important problem for television systems is the jumping or wobbling of the picture due to the movement of the camera. This effect occurs when the camera is mounted in a moving truck or airplane. The Dynalens system has been designed to reduce the effect of rapid scanning motion; see Figure P5.1. A maximum scanning motion of $25^\circ/\text{s}$ is

E5.19 A second-order system has the closed-loop transfer function

$$T(s) = \frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{7}{s^2 + 3.175s + 7}.$$

- (a) Estimate the percent overshoot $P.O.$, the time to peak T_p , and the settling time T_s of the unit step response.
- (b) Obtain the system response to a unit step and verify the results in part (a).

E5.20 Consider the closed-loop system in Figure E5.20, where

$$L(s) = \frac{s+1}{s^2+3s}K_a.$$

- (a) Determine the closed-loop transfer function $T(s) = Y(s)/R(s)$.
- (b) Determine the steady-state error of the closed-loop system response to a unit ramp input.
- (c) Select a value for K_a so that the steady-state error of the system response to a unit step input is zero.

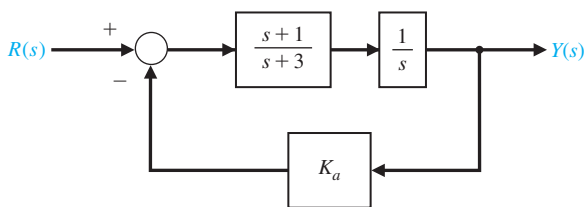


FIGURE E5.20 Nonunity closed-loop feedback control system with parameter K_a .

expected. Let $K_g = K_t = 1$ and assume that τ_g is negligible. (a) Determine the error of the system $E(s)$. (b) Determine the necessary loop gain $K_a K_m K_t$ when a $1^\circ/\text{s}$ steady-state error is allowable. (c) The motor time constant is $\tau_m = 0.40$ s. Determine the necessary loop gain so that the settling time (to within 2% of the final value of v_b) is $T_s \leq 0.03$ s.

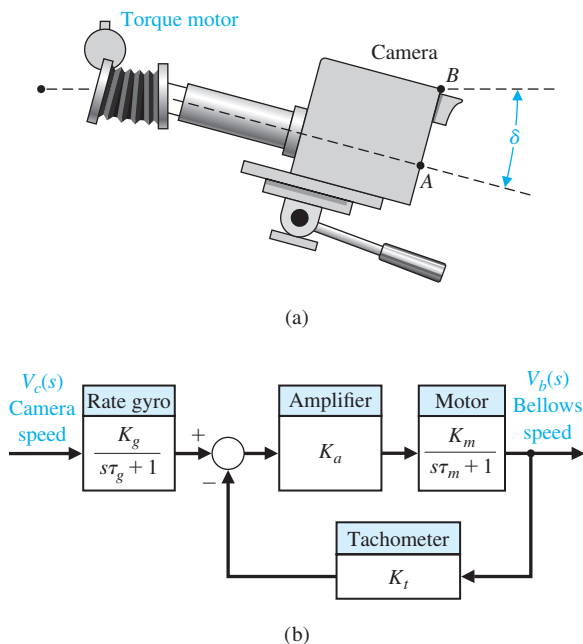


FIGURE P5.1 Camera wobble control.

P5.2 A specific closed-loop control system is to be designed for an underdamped response to a step input. The specifications for the system are as follows:

$$10\% < P.O. < 20\%,$$

$$T_s < 0.6 \text{ s}.$$

(a) Identify the desired area for the dominant roots of the system. (b) Determine the smallest value of a third root r_3 if the complex conjugate roots are to represent the dominant response. (c) The closed-loop system transfer function $T(s)$ is third-order, and the feedback has a unity gain. Determine the loop transfer function $G(s) = Y(s)/E(s)$ when the settling time to within 2% of the final value is $T_s = 0.6 \text{ s}$ and the percent overshoot is $P.O. = 20\%$.

P5.3 A laser beam can be used to weld, drill, etch, cut, and mark metals, as shown in Figure P5.3(a) [14]. Assume we have a work requirement for an accurate laser to mark a parabolic path with a closed-loop control system, as shown in Figure P5.3(b). Calculate the necessary gain to result in a steady-state error of 5 mm for $r(t) = t^2 \text{ cm}$.

P5.4 The loop transfer function of a unity negative feedback system

$$L(s) = G_c(s)G(s) = \frac{K}{s(s+2)}.$$

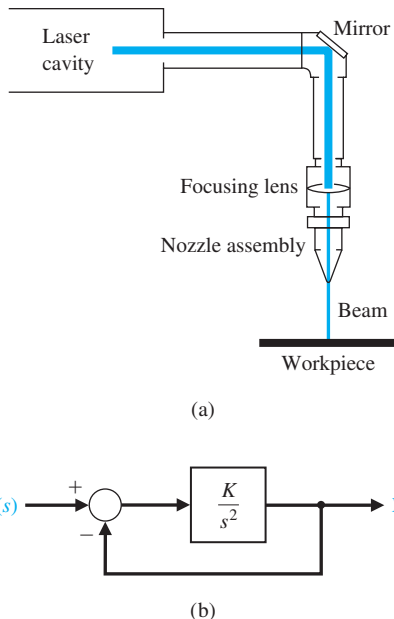


FIGURE P5.3 Laser beam control.

A system response to a step input is specified as follows:

$$T_p = 1.1 \text{ s},$$

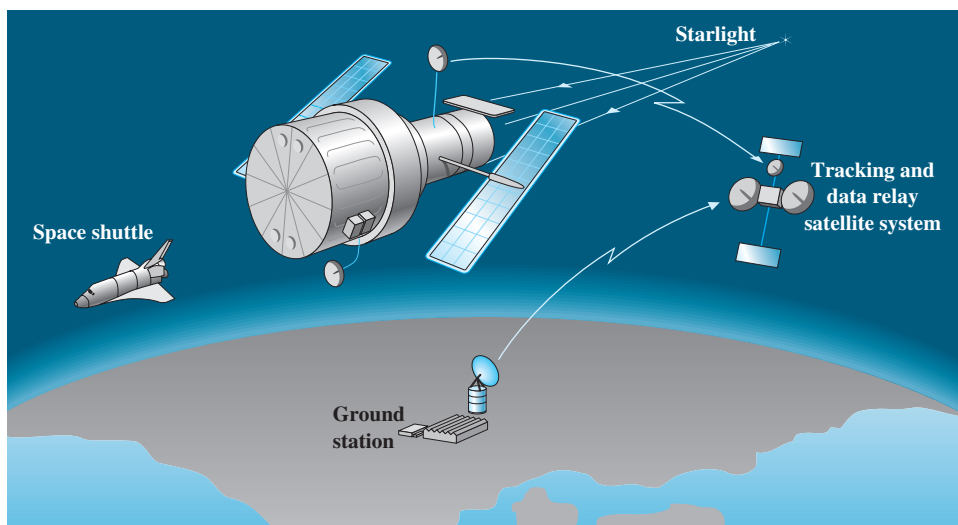
$$P.O. = 5\%.$$

(a) Determine whether both specifications can be met simultaneously. (b) If the specifications cannot be met simultaneously, determine a compromise value for K so that the peak time and percent overshoot specifications are relaxed by the same percentage.

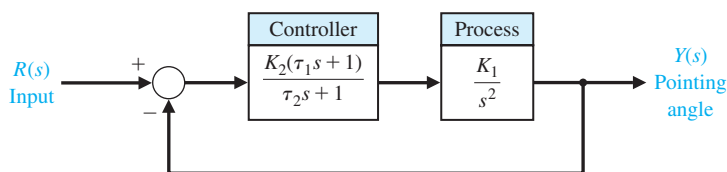
P5.5 A space telescope is to be launched to carry out astronomical experiments [8]. The pointing control system is desired to achieve 0.01 minute of arc and track solar objects with apparent motion up to 0.21 arc minute per second. The system is illustrated in Figure P5.5(a). The control system is shown in Figure P5.5(b). Assume that $\tau_1 = 1 \text{ s}$ and $\tau_2 = 0$. (a) Determine the gain $K = K_1K_2$ required so that the response to a unit step command is as rapid as reasonable with a percent overshoot of $P.O. \leq 5\%$. (b) Determine the steady-state error of the system for a step and a ramp input.

P5.6 A robot is programmed to have a tool or welding torch follow a prescribed path [7, 11]. Consider a robot tool that is to follow a sawtooth path, as shown in Figure P5.6(a). The loop transfer function of the plant is

$$L(s) = G_c(s)G(s) = \frac{75(s+1)}{s(s+5)(s+25)}$$

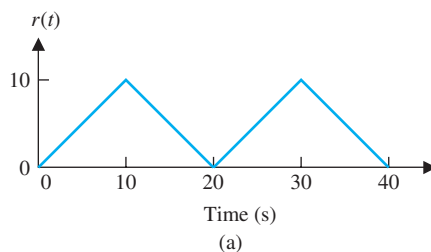


(a)

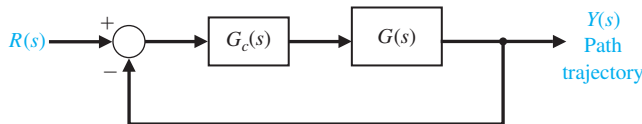


(b)

FIGURE P5.5
(a) The space telescope. (b) The space telescope pointing control system.



(a)



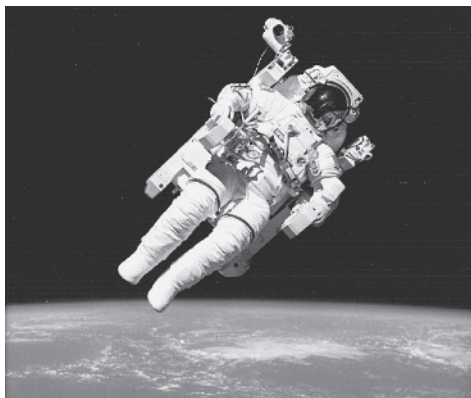
(b)

FIGURE P5.6
Robot path control.

for the closed-loop system shown in Figure 5.6(b). Calculate the steady-state error.

P5.7 Astronaut Bruce McCandless II took the first untethered walk in space on February 7, 1984, using the gas-jet propulsion device illustrated in Figure P5.7(a). The controller can be represented by a gain K_2 , as shown

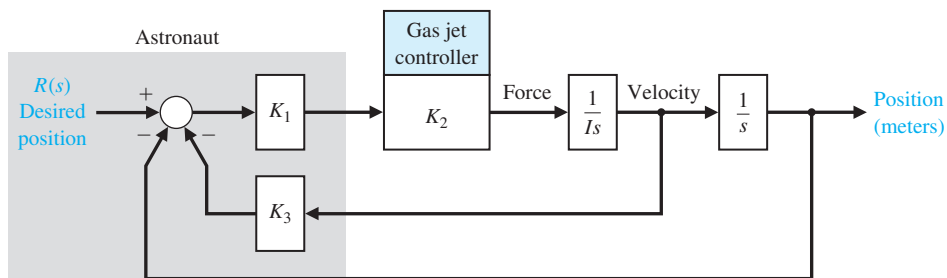
in Figure P5.7(b). The moment of inertia of the equipment and astronaut is $I = 25 \text{ kg m}^2$. (a) Determine the necessary gain K_3 to maintain a steady-state error equal to 1 cm when the input is a unit ramp. (b) With this gain K_3 , determine the necessary gain K_1K_2 in order to restrict the percent overshoot to $P.O. \leq 10\%$.



(a)

FIGURE P5.7

(a) Astronaut Bruce McCandless II is shown a few meters away from the Earth-orbiting space shuttle. He used a nitrogen-propelled hand-controlled device called the manned maneuvering unit. (Courtesy of NASA.) (b) Block diagram.



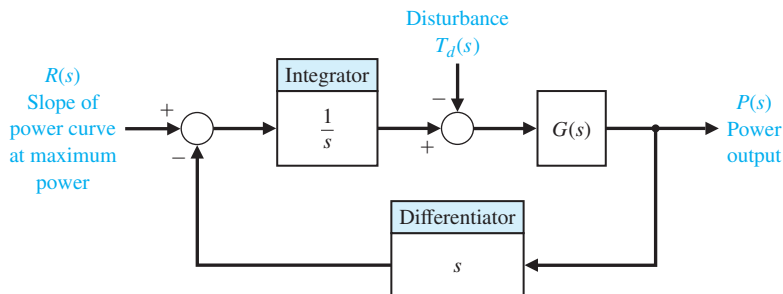
(b)

P5.8 Photovoltaic arrays generate a DC voltage that can be used to drive DC motors or that can be converted to AC power and added to the distribution network. It is desirable to maintain the power out of the array at its maximum available as the solar incidence changes during the day. One such closed-loop system is shown in Figure P5.8. The transfer function for the process is

$$G(s) = \frac{K}{s + 20},$$

where $K = 20$. Find (a) the time constant of the closed-loop system, and (b) the settling time to within 2% of the final value of the system to a unit step disturbance.

P5.9 The antenna that received and transmitted signals to the *Telstar* communication satellite was an extremely large horn antenna. The microwave antenna was 177 ft long, weighed 340 tons, and rolled on a circular track. A photo of the antenna is shown in Figure P5.9. The *Telstar* satellite was 34 inches in diameter and moved

**FIGURE P5.8**

Solar cell control.

about 16,000 mph at an altitude of 2500 miles. The antenna was positioned accurately to 1/10 of a degree, because the microwave beam is 0.2° wide and highly attenuated by the large distance. If the antenna is following the moving satellite, determine the K_v necessary for the system.

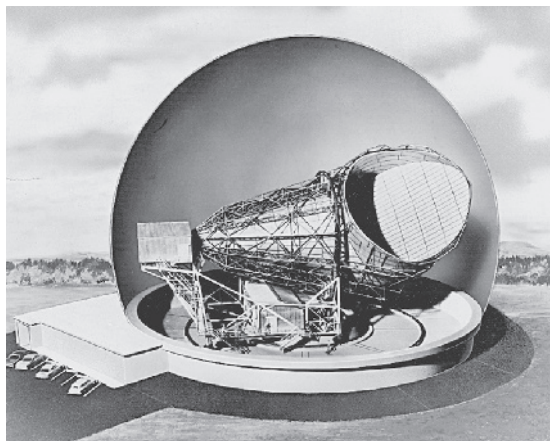


FIGURE P5.9 A model of the antenna for the Telstar System at Andover, Maine. (Photo courtesy of Alcatel-Lucent Technologies.)

P5.10 A speed control system of an armature-controlled DC motor uses the back emf voltage of the motor as a feedback signal. (a) Draw the block diagram of this system (see Example 2.5). (b) Calculate the steady-state error of this system to a step input command setting the speed to a new level. Assume that $R_a = L_a = J = b = 1$, the motor constant is $K_m = 1$, and $K_b = 1$. (c) Select a feedback gain for the back emf signal to yield a step response with a percent overshoot of $P.O. = 15\%$.

P5.11 A unity feedback control system has a process transfer function

$$\frac{Y(s)}{E(s)} = G(s) = \frac{K}{s}.$$

The system input is a step function with an amplitude A . The initial condition of the system at time t_0 is $y(t_0) = Q$, where $y(t)$ is the output of the system. The performance index is defined as

$$I = \int_0^{\infty} e^2(t) dt.$$

(a) Show that $I = (A - Q)^2 / (2K)$. (b) Determine the gain K that will minimize the performance index I . Is this gain a practical value? (c) Select a practical value of gain and determine the resulting value of the performance index.

P5.12 Train travel between cities will increase as trains are developed that travel at high speeds, making the travel time from city center to city center equivalent to airline travel time. The Japanese National Railway has a train called the Shinkansen train that travels an average speed of 320 km/hr [17]. To maintain a desired speed, a speed control system is proposed that yields a zero steady-state error to a ramp input. A third-order system is sufficient. Determine the optimum system transfer function $T(s)$ for an ITAE performance criterion. Estimate the settling time (with a 2% criterion) and percent overshoot for a step input when $\omega_n = 10$.

P5.13 We want to approximate a fourth-order system by a lower-order model. The transfer function of the original system is

$$\begin{aligned} G_H(s) &= \frac{s^3 + 7s^2 + 24s + 24}{s^4 + 10s^3 + 35s^2 + 50s + 24} \\ &= \frac{s^3 + 7s^2 + 24s + 24}{(s+1)(s+2)(s+3)(s+4)}. \end{aligned}$$

Show that if we obtain a second-order model by the method of Section 5.8, and we do not specify the poles and the zero of $G_L(s)$, we have

$$\begin{aligned} G_L(s) &= \frac{0.2917s + 1}{0.399s^2 + 1.375s + 1} \\ &= \frac{0.731(s + 3.428)}{(s + 1.043)(s + 2.4)}. \end{aligned}$$

P5.14 For the original system of Problem P5.13, we want to find the lower-order model when the poles of the second-order model are specified as -1 and -2 and the model has one unspecified zero. Show that this low-order model is

$$G_L(s) = \frac{0.986s + 2}{s^2 + 3s + 2} = \frac{0.986(s + 2.028)}{(s+1)(s+2)}.$$

P5.15 Consider a unity feedback system with loop transfer function

$$L(s) = G_c(s)G(s) = \frac{K(s+2)}{(s+5)(s^2+s+10)}.$$

Determine the value of the gain K such that the percent overshoot to a unit step is minimized.

P5.16 A magnetic amplifier with a low-output impedance is shown in Figure P5.16 in cascade with a low-pass filter and a preamplifier. The amplifier has a high-input impedance and a gain of 1 and is used for adding the signals as shown. Select a value for the capacitance C so that the transfer function $V_0(s)/V_{in}(s)$ has a damping ratio of $1/\sqrt{2}$. The time constant of the magnetic amplifier is equal to 1 second, and the gain is $K = 10$. Calculate the settling time (with a 2% criterion) of the resulting system.

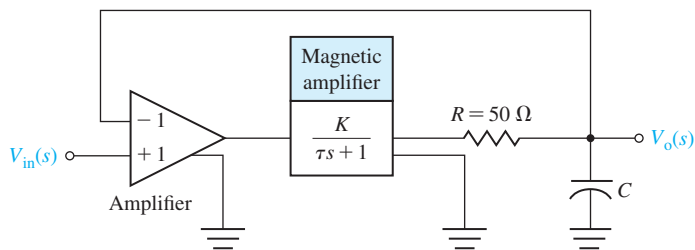


FIGURE P5.16
Feedback amplifier.

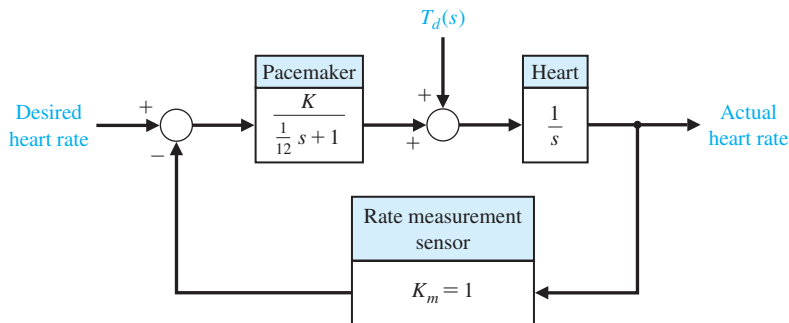


FIGURE P5.17
Heart pacemaker.

P5.17 Electronic pacemakers for human hearts regulate the speed of the heart pump. A proposed closed-loop system that includes a pacemaker and the measurement of the heart rate is shown in Figure P5.17 [2, 3]. The transfer function of the heart pump and the pacemaker is found to be

$$G(s) = \frac{K}{s(s/12 + 1)}.$$

Design the amplifier gain to yield a system with a settling time to a step disturbance of less than 1 second. The percent overshoot to a step in desired heart rate should be $P.O. \leq 10\%$. (a) Find a suitable range of K . (b) If the nominal value of K is $K = 10$, find the sensitivity of the system to small changes in K . (c) Evaluate the sensitivity of part (b) at DC (set $s = 0$). (d) Evaluate the magnitude of the sensitivity at the normal heart rate of 60 beats/minute.

P5.18 Consider the original third-order system given in Example 5.7. Determine a first-order model with one pole unspecified and no zeros that will represent the third-order system.

P5.19 A closed-loop control system with negative unity feedback has a loop transfer function

$$L(s) = G_c(s)G(s) = \frac{8}{s(s^2 + 6s + 12)}.$$

- Determine the closed-loop transfer function $T(s)$.
- Determine a second-order approximation for $T(s)$.
- Plot the response of $T(s)$ and the second-order approximation to a unit step input and compare the results.

P5.20 A system is shown in Figure P5.20.

- Determine the steady-state error for a unit step input in terms of K and K_1 , where $E(s) = R(s) - Y(s)$.
- Select K_1 so that the steady-state error is zero.

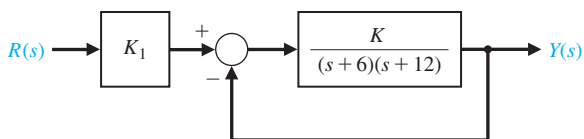


FIGURE P5.20 System with pregain, K_1 .

P5.21 Consider the closed-loop system in Figure P5.21. Determine values of the parameters k and a so that the following specifications are satisfied:

- The steady-state error to a unit step input is zero.
- The closed-loop system has a percent overshoot of $P.O. \leq 5\%$.

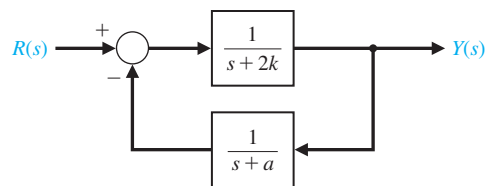


FIGURE P5.21 Closed-loop system with parameters k and a .

P5.22 Consider the closed-loop system in Figure P5.22, where

$$G_c(s)G(s) = \frac{2}{s + 0.2K} \quad \text{and} \quad H(s) = \frac{2}{2s + \tau}.$$

- If $\tau = 2.43$, determine the value of K such that the steady-state error of the closed-loop system response to a unit step input, is zero.
- Determine the percent overshoot and the time to peak of the unit step response when K is as in part (a).

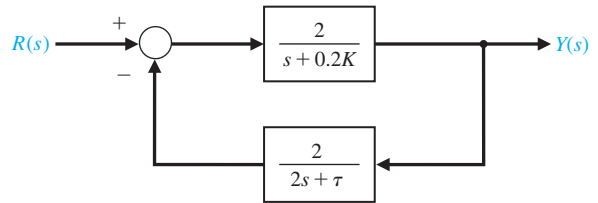


FIGURE P5.22 Nonunity closed-loop feedback control system.

ADVANCED PROBLEMS

AP5.1 A closed-loop transfer function is

$$T(s) = \frac{Y(s)}{R(s)} = \frac{108(s + 3)}{(s + 9)(s^2 + 8s + 36)}.$$

- Determine the steady-state error for a unit step input.
- Assume that the complex poles dominate, and determine the percent overshoot and settling time to within 2% of the final value.
- Plot the actual system response, and compare it with the estimates of part (b).

AP5.2 A closed-loop system is shown in Figure AP5.2. Plot the response to a unit step input for the system for $\tau_z = 0, 0.05, 0.1$, and 0.5 . Record the percent overshoot, rise time, and settling time (with a 2% criterion) as τ_z varies. Describe the effect of varying τ_z . Compare the location of the zero $-1/\tau_z$ with the location of the closed-loop poles.

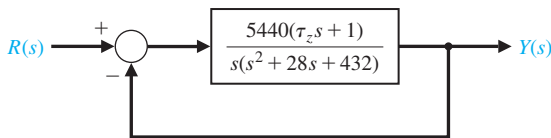


FIGURE AP5.2 System with a variable zero.

AP5.3 A closed-loop system is shown in Figure AP5.3. Plot the response to a unit step input for the system with $\tau_p = 0, 0.5, 2$, and 5 . Record the percent overshoot, rise time, and settling time (with a 2% criterion) as τ_p varies. Describe the effect of varying τ_p . Compare the location of the open-loop pole $-1/\tau_p$ with the location of the closed-loop poles.

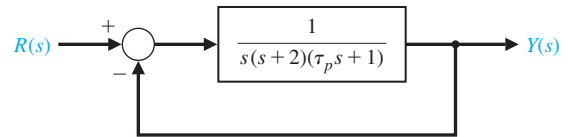


FIGURE AP5.3 System with a variable pole in the process.

AP5.4 The speed control of a high-speed train is represented by the system shown in Figure AP5.4 [17]. Determine the equation for steady-state error for K for a unit step input. Consider the three values for K equal to 1, 10, and 100.

- Determine the steady-state error.
- Determine and plot the response $y(t)$ for (i) a unit step input $R(s) = 1/s$ and (ii) a unit step disturbance input $T_d(s) = 1/s$.
- Create a table showing percent overshoot, settling time (with a 2% criterion), e_{ss} for $r(t)$, and $|y/t_d|_{\max}$ for the three values of K . Select the best compromise value.

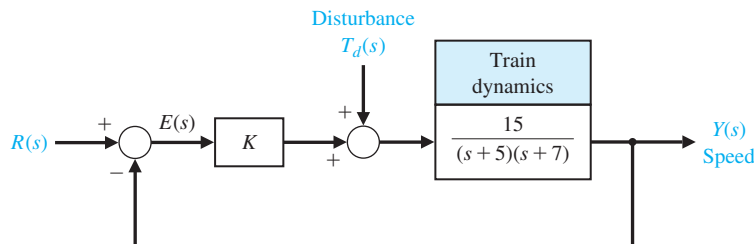


FIGURE AP5.4 Speed control.

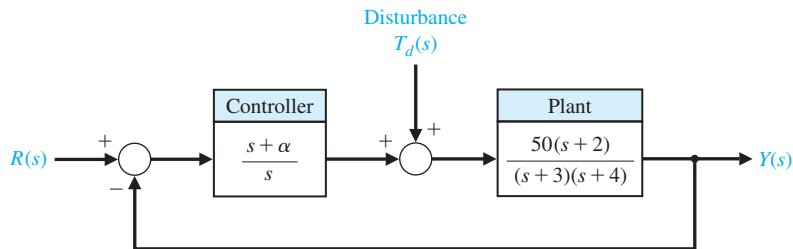


FIGURE AP5.5
System with control parameter α .

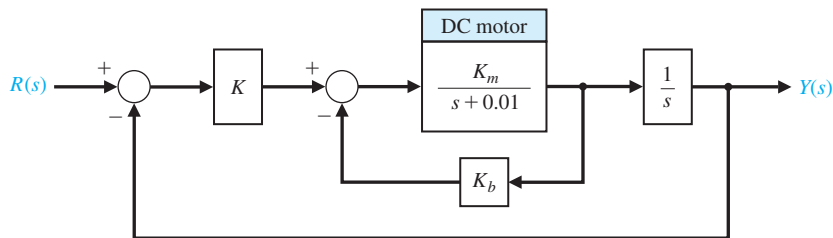


FIGURE AP5.6
DC motor control.

AP5.5 A system with a controller is shown in Figure AP5.5. The zero of the controller may be varied. Let $\alpha = 0, 10, 100$.

- Determine the steady-state error for a unit step input for $\alpha = 0$ and $\alpha \neq 0$.
- Plot the response of the system to a unit step input disturbance for the three values of α . Compare the results and select the best value of the three values of α .

AP5.6 The block diagram model of an armature-current-controlled DC motor is shown in Figure AP5.6.

- Determine the steady-state tracking error to a ramp input in terms of K , K_b , and K_m .
- Let $K_m = 10$ and $K_b = 0.05$, and select K so that steady-state tracking error is equal to 1.
- Plot the response to a unit step input and a unit ramp input for 20 seconds. Are the responses acceptable?

AP5.7 Consider the closed-loop system in Figure AP5.7 with transfer functions

$$G_c(s) = \frac{100}{s + 100} \quad \text{and} \quad G(s) = \frac{K}{s(s + 50)},$$

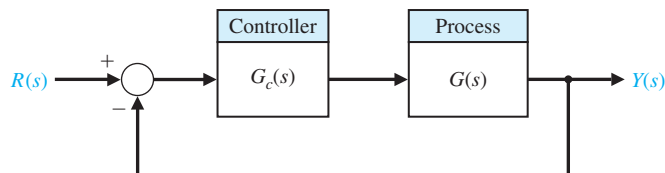


FIGURE AP5.7
Closed-loop system with unity feedback.

where

$$1000 \leq K \leq 5000.$$

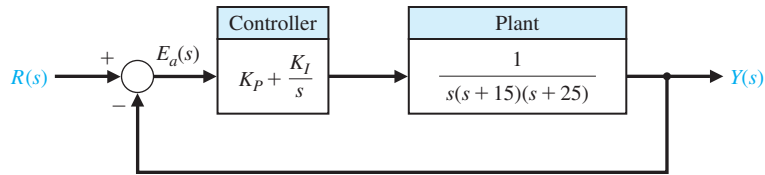
- Assume that the complex poles dominate and estimate the settling time and percent overshoot to a unit step input for $K = 1000, 2000, 3000, 4000$, and 5000 .
- Determine the actual settling time and percent overshoot to a unit step for the values of K in part (a).
- Co-plot the results of (a) and (b) and comment.

AP5.8 A unity negative feedback system has the loop transfer function

$$L(s) = G_c(s)G(s) = \frac{K(s + 2)}{s^2 + \frac{2}{3}s + \frac{1}{3}}.$$

Determine the gain K that minimizes the damping ratio ζ of the closed-loop system poles. What is the minimum damping ratio?

FIGURE AP5.9
Feedback control system with a proportional plus integral controller.



AP5.9 The unity negative feedback system in Figure AP5.9 has the process given by

$$G(s) = \frac{1}{s(s+15)(s+25)}.$$

The controller is a proportional plus integral controller with gains K_p and K_I . The objective is to design the controller gains such that the dominant roots have a damping ratio ζ equal to 0.707. Determine the resulting peak time and settling time (with a 2% criterion) of the system to a unit step input.

DESIGN PROBLEMS

CDP5.1 The capstan drive system of the previous problems (see CDP1.1–CDP4.1) has a disturbance due to changes in the part that is being machined as material is removed. The controller is an amplifier $G_c(s) = K_a$. Evaluate the effect of a unit step disturbance, and determine the best value of the amplifier gain so that the percent overshoot to a step command $r(t) = A, t > 0$ is $P.O. \leq 5\%$, while reducing the effect of the disturbance as much as possible.

DP5.1 The roll control autopilot of an aircraft is shown in Figure DP5.1. The goal is to select a suitable K so that the response to a step command $\phi_d(t) = A, t \geq 0$, will provide a response $\phi(t)$ that is a fast response and has a percent overshoot of $P.O. \leq 20\%$. (a) Determine the closed-loop transfer function $\phi(s)/\phi_d(s)$. (b) Determine the roots of the characteristic equation for $K = 0.7, 3$, and 6.

(c) Using the concept of dominant roots, find the expected percent overshoot and peak time for the approximate second-order system. (d) Plot the actual response and compare with the approximate results of part (c). (e) Select the gain K so that the percent overshoot is $P.O. = 16\%$. What is the resulting peak time?

DP5.2 The design of the control for a welding arm with a long reach requires the careful selection of the parameters [13]. The system is shown in Figure DP5.2, where $\zeta = 0.6$, and the gain K and the natural frequency ω_n can be selected. (a) Determine K and ω_n so that the response to a unit step input achieves a peak time for the first overshoot (above the desired level of 1) that is $T_p \leq 1$ s and the percent overshoot is $P.O. \leq 5\%$. (b) Plot the response of the system designed in part (a) to a step input.

FIGURE DP5.1
Roll angle control.

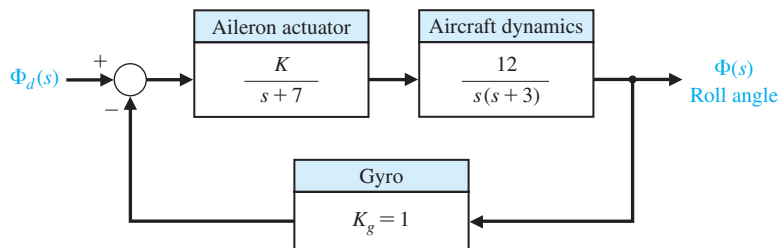


FIGURE DP5.2
Welding tip position control.

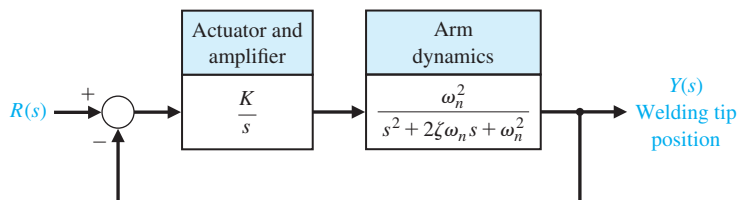
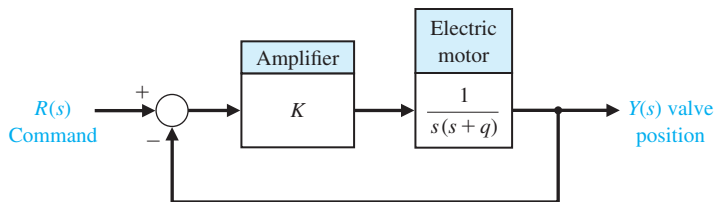


FIGURE DP5.3
Active suspension system.



DP5.3 Active suspension systems for modern automobiles provide a comfortable firm ride. The design of an active suspension system adjusts the valves of the shock absorber so that the ride fits the conditions. A small electric motor, as shown in Figure DP5.3, changes the valve settings [13]. Select a design value for K and the parameter q in order to satisfy the ITAE performance for a step command $R(s)$ and a settling time (with a 2% criterion) for the step response of $T_s \leq 0.5$ s. Upon completion of your design, predict the resulting percent overshoot for a step input.

DP5.4 The space satellite shown in Figure DP5.4(a) uses a control system to readjust its orientation, as shown in Figure DP5.4(b).

- Determine a second-order model for the closed-loop system.
- Using the second-order model, select a gain K so that the percent overshoot is $P.O. \leq 15\%$ and the steady-state error to a step is less than 12%.

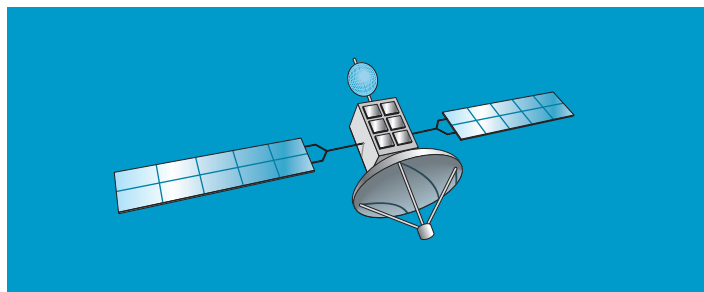
(c) Verify your design by determining the actual performance of the third-order system.

DP5.5 A deburring robot can be used to smooth off machined parts by following a preplanned path (input command signal). In practice, errors occur due to robot inaccuracy, machining errors, large tolerances, and tool wear. These errors can be eliminated using force feedback to modify the path online [8, 11].

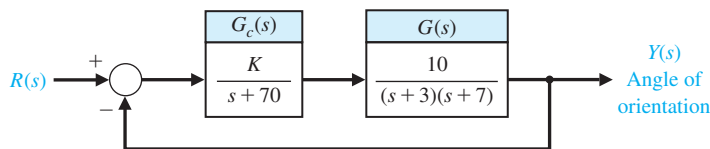
While force control has been able to address the problem of accuracy, it has been more difficult to solve the contact stability problem. In fact, by closing the force loop and introducing a compliant wrist force sensor (the most common type of force control), one can add to the stability problem.

A model of a robot deburring system is shown in Figure DP5.5. Determine the region of stability for the system for K_1 and K_2 . Assume both adjustable gains are greater than zero.

DP5.6 The model for a position control system using a DC motor is shown in Figure DP5.6. The goal is to select K_1 and K_2 so that the peak time is $T_p \leq 0.5$ s



(a)



(b)

FIGURE DP5.4
Control of a space satellite.

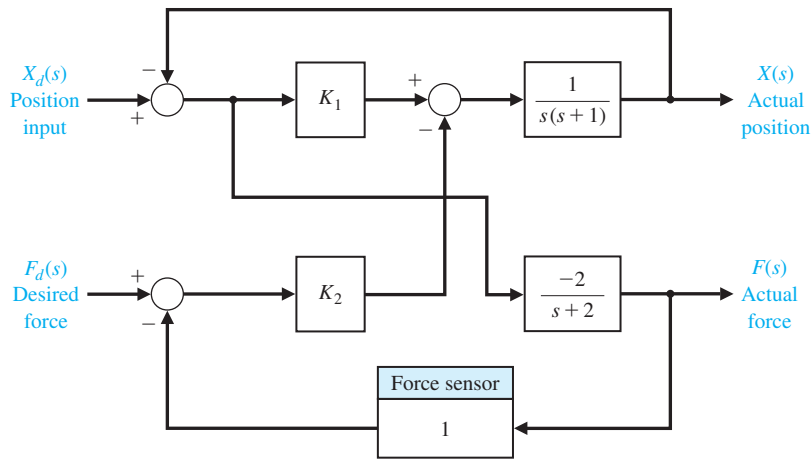


FIGURE DP5.5
Deburring robot.

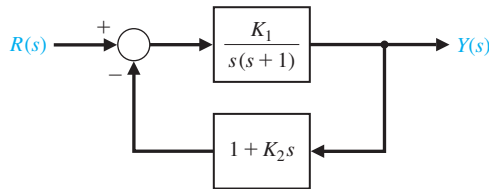


FIGURE DP5.6
Position control robot.

and the percent overshoot $P.O.$ for a step input is $P.O. \leq 2\%$.

DP5.7 A three-dimensional cam for generating a function of two variables is shown in Figure DP5.7(a). Both x and y may be controlled using a position control system [31]. The control of x may be achieved with a DC motor and position feedback of the form shown in Figure DP5.7(b), with the DC motor and load represented by

$$G(s) = \frac{K}{s(s+p)(s+4)},$$

where $K = 2$ and $p = 2$. Design a proportional plus derivative controller

$$G_c(s) = K_p + K_D s$$

to achieve a percent overshoot $P.O. \leq 5\%$ to a unit step input and a settling time $T_s \leq 2$ s.

DP5.8 Computer control of a robot to spray-paint an automobile is accomplished by the system shown in Figure DP5.8(a) [7]. We wish to investigate the system when $K = 1, 10$, and 20 . The feedback control

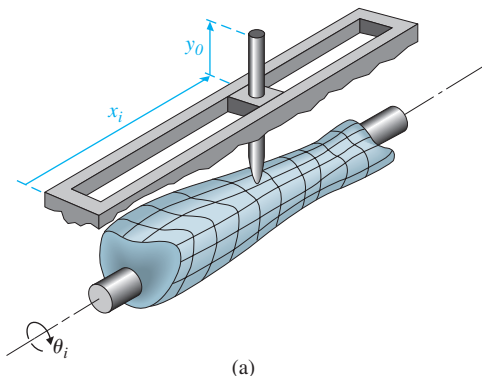
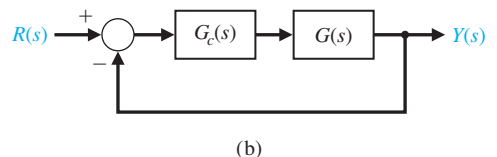


FIGURE DP5.7
(a) Three-dimensional cam and (b) x -axis control system.



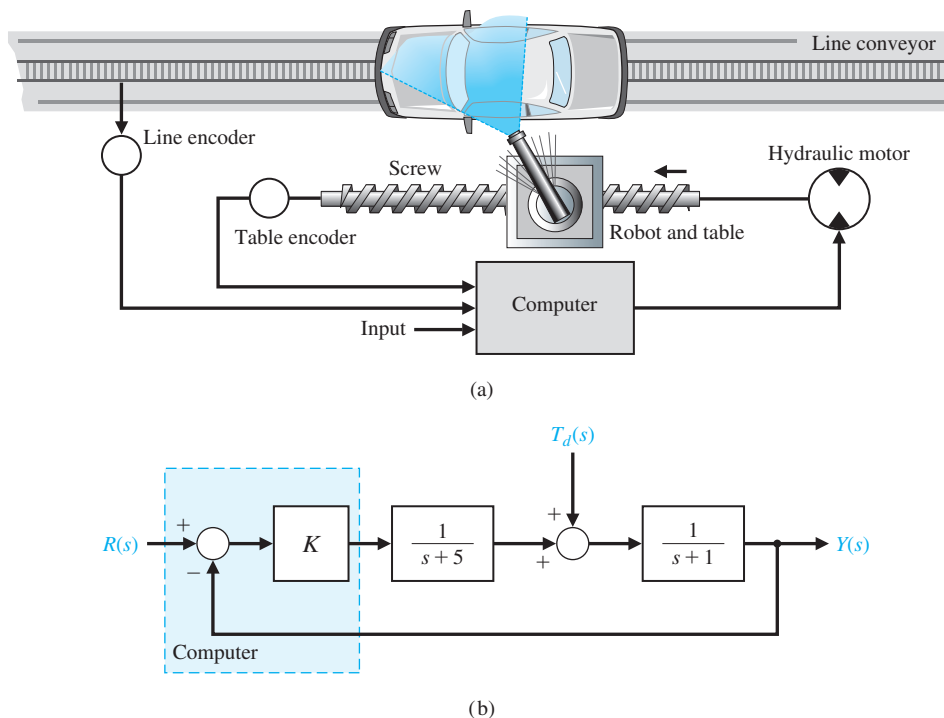


FIGURE DP5.8
Spray-paint robot.

block diagram is shown in Figure DP5.8(b). (a) For the three values of K , determine the percent overshoot, the settling time (with a 2% criterion), and the steady-state error for a unit step input. Record your results in

a table. (b) Choose one of the three values of K that provides acceptable performance. (c) For the value selected in part (b), determine the output for a disturbance $T_d(s) = 1/s$ when $R(s) = 0$.



COMPUTER PROBLEMS

CP5.1 Consider the closed-loop transfer function

$$T(s) = \frac{15}{s^2 + 8s + 15}.$$

Obtain the impulse response analytically and compare the result to one obtained using the impulse function.

CP5.2 A unity negative feedback system has the loop transfer function

$$L(s) = G_c(s)G(s) = \frac{s + 10}{s^2(s + 15)}.$$

Using Isim, obtain the response of the closed-loop system to a unit ramp input,

$$R(s) = 1/s^2.$$

Consider the time interval $0 \leq t \leq 50$. What is the steady-state error?

CP5.3 A working knowledge of the relationship between the pole locations of the second-order system shown in Figure CP5.3 and the transient response is important in control design. With that in mind, consider the following four cases:

1. $\omega_n = 2, \quad \zeta = 0,$
2. $\omega_n = 2, \quad \zeta = 0.1,$
3. $\omega_n = 1, \quad \zeta = 0,$
4. $\omega_n = 1, \quad \zeta = 0.2.$

Using the impulse and subplot functions, create a plot containing four subplots, with each subplot depicting the impulse response of one of the four cases listed. Compare the plot with Figure 5.17 in Section 5.5, and discuss the results.

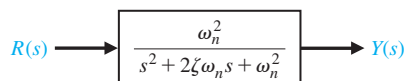
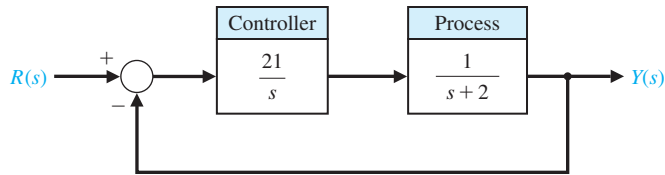


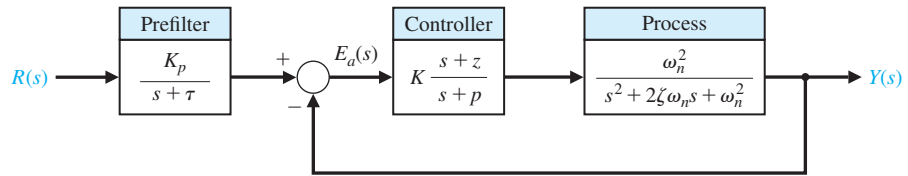
FIGURE CP5.3 A simple second-order system.

FIGURE CP5.4

A negative feedback control system.

**FIGURE CP5.5**

Feedback control system with controller and prefilter.



CP5.4 Consider the control system shown in Figure CP5.4.

- Show analytically that the expected percent overshoot of the closed-loop system response to a unit step input is $P.O. = 50\%$.
- Develop an m-file to plot the unit step response of the closed-loop system and estimate the percent overshoot from the plot. Compare the result with part (a).

CP5.5 Consider the feedback system in Figure CP5.5. Develop an m-file to design a controller and prefilter

$$G_c(s) = K \frac{s+z}{s+p} \quad \text{and} \quad G_p(s) = \frac{K_p}{s+\tau}$$

such that the ITAE performance criterion is minimized. For $\omega_n = 0.45$ and $\zeta = 0.59$, plot the unit step response and determine the percent overshoot and settling time.

CP5.6 The loop transfer function of a unity negative feedback system is

$$L(s) = G_c(s)G(s) = \frac{25}{s(s+5)}.$$

Develop an m-file to plot the unit step response and determine the values of peak overshoot M_p , time to peak T_p , and settling time T_s (with a 2% criterion).

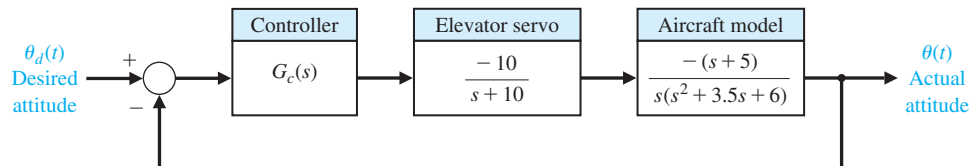
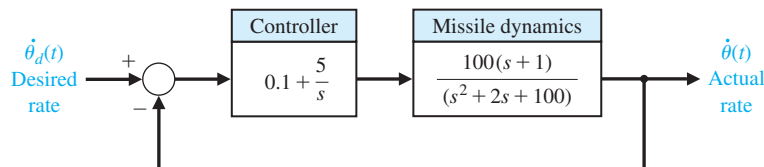
CP5.7 An autopilot designed to hold an aircraft in straight and level flight is shown in Figure CP5.7.

- Suppose the controller is a constant gain controller given by $G_c(s) = 2$. Using the `lsim` function, compute and plot the ramp response for $\theta_d(t) = at$, where $a = 0.5^\circ/\text{s}$. Determine the attitude error after 10 seconds.
- If we increase the complexity of the controller, we can reduce the steady-state tracking error. With this objective in mind, suppose we replace the constant gain controller with the more sophisticated controller

$$G_c(s) = K_1 + \frac{K_2}{s} = 2 + \frac{1}{s}.$$

This type of controller is known as a proportional plus integral (PI) controller. Repeat the simulation of part (a) with the PI controller, and compare the steady-state tracking errors of the constant gain controller versus the PI controller.

CP5.8 The block diagram of a rate loop for a missile autopilot is shown in Figure CP5.8. Using the analytic formulas for second-order systems, predict M_p , T_p , and T_s for the closed-loop system due to a unit step input. Compare the predicted results with the actual

FIGURE CP5.7
An aircraft autopilot block diagram.**FIGURE CP5.8**
A missile rate loop autopilot.

unit step response obtained with the **step** function. Explain any differences.

CP5.9 Develop an m-file that can be used to analyze the closed-loop system in Figure CP5.9. Drive the system with a step input and display the output on a graph. What is the settling time and the percent overshoot?

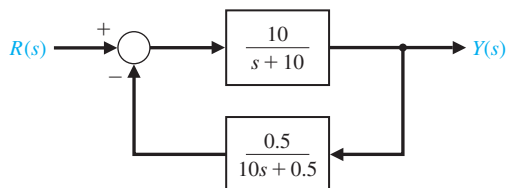


FIGURE CP5.9 Nonunity feedback system.

CP5.10 Develop an m-file to simulate the response of the system in Figure CP5.10 to a ramp input $R(s) = 1/s^2$. What is the steady-state error? Display the output on an x - y graph.

CP5.11 Consider the closed-loop system in Figure CP5.11. Develop an m-file to accomplish the following tasks:

- Determine the closed-loop transfer function $T(s) = Y(s)/R(s)$.

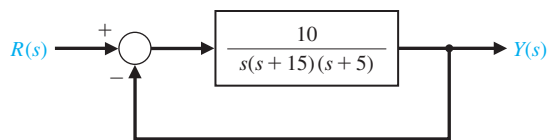


FIGURE CP5.10 Closed-loop system for m-file.

- Plot the closed-loop system response to an impulse input $R(s) = 1$, a unit step input $R(s) = 1/s$, and a unit ramp input $R(s) = 1/s^2$. Use the **subplot** function to display the three system responses.

CP5.12 A closed-loop transfer function is given by

$$T(s) = \frac{Y(s)}{R(s)} = \frac{77(s+2)}{(s+7)(s^2+4s+22)}.$$

- Obtain the response of the closed-loop transfer function $T(s) = Y(s)/R(s)$ to a unit step input. What is the settling time T_s (use a 2% criterion) and percent overshoot $P.O.$?
- Neglecting the real pole at $s = -7$, determine the settling time T_s and percent overshoot $P.O.$. Compare the results with the actual system response in part (a). What conclusions can be made regarding neglecting the pole?

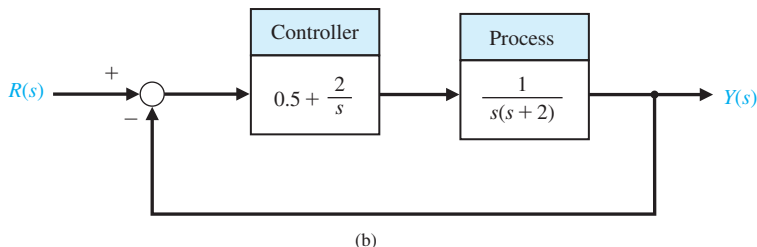
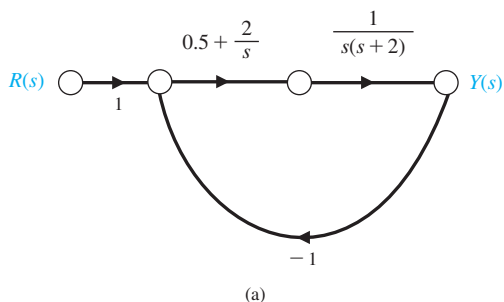


FIGURE CP5.11 A single loop unity feedback system. (a) Signal flow graph. (b) Block diagram.

ANSWERS TO SKILLS CHECK

True or False: (1) True; (2) False; (3) False; (4) True; (5) False

Multiple Choice: (6) a; (7) a; (8) c; (9) b; (10) b; (11) a; (12) b; (13) b; (14) a; (15) b

Word Match (in order, top to bottom): i, j, d, g, k, c, n, p, o, b, e, l, f, h, m, a

TERMS AND CONCEPTS

Acceleration error constant, K_a The constant evaluated as $\lim_{s \rightarrow 0} [s^2 G_c(s)G(s)]$. The steady-state error for a parabolic input, $r(t) = At^2/2$, is equal to A/K_a .

Design specifications A set of prescribed performance criteria.

Dominant roots The roots of the characteristic equation that cause the dominant transient response of the system.

Optimum control system A system whose parameters are adjusted so that the performance index reaches an extremum value.

Peak time The time for a system to respond to a step input and rise to a peak response.

Percent overshoot The amount by which the system output response proceeds beyond the desired response.

Performance index A quantitative measure of the performance of a system.

Position error constant, K_p The constant evaluated as $\lim_{s \rightarrow 0} G_c(s)G(s)$. The steady-state error for a step input (of magnitude A) is equal to $A/(1 + K_p)$.

Rise time The time for a system to respond to a step input and attain a response equal to a percentage of the magnitude of the input. The 0–100% rise time, T_r ,

measures the time to 100% of the magnitude of the input. Alternatively, T_{r1} measures the time from 10% to 90% of the response to the step input.

Settling time The time required for the system output to settle within a certain percentage of the input amplitude.

Steady-state response The constituent of the system response that exists a long time following any signal initiation.

Test input signal An input signal used as a standard test of a system's ability to respond adequately.

Transient response The constituent of the system response that disappears with time.

Type number The number N of poles of the transfer function, $G_c(s)G(s)$, at the origin. $G_c(s)G(s)$ is the loop transfer function.

Unit impulse A test input consisting of an impulse of infinite amplitude and zero width, and having an area of unity. The unit impulse is used to determine the impulse response.

Velocity error constant, K_v The constant evaluated as $\lim_{s \rightarrow 0} [sG_c(s)G(s)]$. The steady-state error for a ramp input (of slope A) for a system is equal to A/K_v .